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Group 6

**Optimising Budget Allocation for Reforestation and
Biodiversity: A Mathematical Programming Approach
Municipality-Level Conservation in Pará, Brazil.**

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1 Introduction

SDG Climate and biodiversity action has emerged as a critical challenge of the 21st century - one of the key transitions required to achieve the Sustainable Development Goals (SDGs) by 2030. Forests are critical to this, acting simultaneously as climate regulators and drivers of biodiversity while being casualties of both [1]. This project addresses SDG 15: "life on land", specifically target 15.2 - promoting sustainable forest management, halting deforestation, and increasing reforestation globally [2], examined through the Amazon Rainforest in Pará, Brazil. These objectives are examined through the lens of reforestation efforts in the Amazon Rainforest, specifically within the state of Pará, Brazil. This ecological cornerstone is approaching a critical point of no return. Current estimates show that 17% of the Amazon Rainforest has already been cleared [3]. If deforestation surpasses the 20-25% threshold, the rainforest may incur habitat loss, increased wildfires, and possible transformation into a degraded, dry savanna due to a breakdown in rainfall generation [4].

General Rainforest The Amazon Rainforest plays a crucial role in shaping the planet's future climate and biodiversity strategies, capturing ≈ 2 billion tons of carbon dioxide (CO_2) [5] and containing a third of the world's tropical rainforests [6] with 75% being endemic [7]. Of these species, 95% have experienced habitat loss due to deforestation and its secondary consequences, e.g. increased risk of forest fires and droughts [8]. Additionally, 5% – 9% of Amazon's vascular plant species are projected to become extinct by 2050 due to further habitat loss [9].

Presidential Policies The necessity of sustained, state and federal-level commitment is evident from the historical volatility of deforestation levels in Brazil [10, 11]. Under Jair Bolsonaro's administration (2019-2022), the power and autonomy of forest protection agencies were severely diminished. This led to a 60% increase in deforestation compared to the previous presidential term [12]. With Luiz Inacio Lula da Silva's re-election in 2023, environmental policy has been reinstated and prioritised. The return of policy enforcement contributed directly to a 43% reduction in new deforestation within the first six months of his term [13]. As defined under PPCDAm (2023-2027), zero deforestation explicitly mandates the recovery of native vegetation through financial distribution and sustainable management [14]. This restoration mandate aligns with Brazil's Nationally Determined Contribution (NDC) to restore 12 million hectares of deforested rainforest land by 2030 [15]. Given the capital-intensive nature of reforestation, meeting Brazil's NDC is estimated to cost USD \$15.6 billion [16].

Pará Context Aligning with the national PPCDAm framework, Pará operates under the Amazônia Agora State Plan (PEAA). The PEAA sets a binding target of a 37% reduction in GHG emissions from deforestation by 2030, and the regeneration of 5.65 million hectares of native vegetation, 47% of Brazil's total commitment under the NDC [17]. These targets are not aspirational, as the state is responsible for 37% of the deforestation in the Brazilian Amazon, making it the most critical sub-national jurisdiction for meeting Brazil's NDC commitments [18]. However, applying Brancalion et al.'s [16] per-hectare cost range to PEAA's target yields a total commitment of USD \$7.4 billion.

Decision Maker Responsibility for implementing the PEAA rests with the Secretaria de Estado de Meio Ambiente e Sustentabilidade (SEMAs) [19]. As the principal environmental institution within the state government, SEMAs plays a central role in translating environmental policy objectives into operational programmes.

SEMAs functions as the central decision-maker, responsible for allocating all state-level resources to reforestation and restoration initiatives across Pará's 144 municipalities [19]. Beyond the management of state funds, SEMAs interfaces with external and private climate finance and restoration investment initiatives. While SEMAs does not fully control private or international capital flows, its regulatory authority and project approval processes provide significant influence over where restoration investments are deployed [20]. In practice, investors partner with SEMAs to deploy funds, as evidenced by its coordinating role in the Jurisdictional REDD+ System [21, 22]. Therefore, for this project, SEMAs is assumed to act as the central coordinating authority responsible for allocating both state and external funds.

Problem outline This report estimates that SEMAs has access to USD \$489.85 million per year (*B*) [Section 5.1.1] that it can allocate to reforestation initiatives across Pará's 144 municipalities. However, *B* falls short of the estimated USD \$740 million annual commitment required to meet the PEAA target, accounting for only 66.20% of the total need. Moreover, SEMAs's role as a central allocator is complicated due to the extreme socio-ecological heterogeneity of Pará's 144 municipalities, resulting in reforestation opportunities differing substantially in ecological return, implementation cost and the risk of efforts being reversed [19]. SEMAs therefore faces competing priorities, framing reforestation as a constrained resource allocation problem requiring formal optimisation.

Structure This report contains five subsequent sections: Section 2 presents literature that examines the efficacy of linear optimisation in relevant contexts, providing the foundations of this report. Section 3 discusses the details of the models and presents the key trade-offs that must be navigated. Section 4 presents the mathematical formulation of the deterministic and stochastic models, outlining the decision variables, parameters and constraints. Section 5 outlines the data that this report used, along with its computational results; Section 6 concludes and makes recommendations for SEMAs.

2 Literature Review

The literature surrounding site optimisation using mathematical programming, both in the scope of reforestation and beyond, is wide-ranging:

This reforestation project undertakes the structural characteristics of Systematic Conservation Planning (SCP), as defined by Beyer et al. [23]. The quality of selecting any planning allocations depends on their complementarity with other units. Hence, ranking or greedy algorithms are inadequate for determining efficient allocations. Traditionally, Simulated Annealing (SA) algorithms were implemented for SCP. However, Beyer et al. demonstrate that these produce solutions of unknown quality, with no guarantee of proximity to the global optimum. In their benchmarking experiments, SA required approximately 14 hours to achieve a solution within 19% of optimality for a moderately sized quadratic problem, whereas Integer Linear Programming (ILP) achieved solutions within 0.5% of optimality in under 30 seconds using Gurobi. Such advances make ILP the superior choice for linear and quadratic SCP problems, also enabling rigorous trade-off analysis across competing objectives. This project extends the literature's ILP framework by adopting a Mixed-Integer Linear Program (MILP). Following Beyer et al.'s logic that the quality of any heuristic solution can only be assessed relative to an exact optimum, a constraint-aware greedy heuristic was implemented as a benchmark to allow an optimality gap to be empirically quantified and justify the efficiency gains of the optimisation approach.

Orsi et al. [24] developed an ILP model to prioritise forest landscape restoration in Chiapas, Mexico, integrating spatial optimisation with ecological criteria to identify reforestation sites that maximise conservation value under fixed budget constraints. By deriving biodiversity solely from tree species richness, the model adopts a relatively narrow definition of biodiversity, excluding other organisms such as fauna, fungi, and microorganisms. These species play a critical role in ecosystem recovery through processes such as seed dispersal, pollination, and nutrient cycling. This limitation is particularly relevant in the context of Pará, where reforestation depends on all species present in the Amazon rainforest. While our model draws on the biodiversity-oriented optimisation framework proposed by Orsi et al., it addresses a fundamentally different decision problem. Rather than focusing solely on spatial prioritisation of restoration sites based on tree species distributions, our model formulates reforestation as an allocation problem at the municipality level, integrating floral and faunal biodiversity, deforestation urgency, restoration cost, extent of degradation, and reversal risk.

Assunção et al. [25] developed an ILP model to optimise the "Priority List" of municipalities in the Brazilian Amazon, aiming to minimise total deforestation and carbon emissions. While the government utilised a retrospective threshold rule to populate the "priority list", targeting municipalities that exceeded specific historical deforestation levels, Assunção et al. optimised selection based on predicted future impact. Utilising a minimax approach to account for partial identification, their model chose a list to minimise the worst-case deforestation scenario, reducing carbon emissions by over 10% compared to the actual list. However, similar to Cocks and Baird, this model treats municipality selection as a binary variable, failing to capture the extent or intensity of intervention through a continuous variable. Furthermore, Assunção et al.'s framework determines where to apply regulatory pressure instead of addressing the allocation of

scarce financial resources for proactive ecological recovery. It emphasises enforcement over investment in restoration. By incorporating a continuous funding variable alongside binary selection, our model broadens this framework to optimise not only which municipalities to prioritise but also the scale of reforestation investment within each.

Weiskopf et al. (2026) [26] develop an ILP restoration-prioritisation framework where restoration demand exceeds budget and delivery capacity. Benefits are constructed from predicted reductions in extinction risk, combined with cost information under feasibility constraints, a cap on total restored area, and minimum restoration requirements across habitat types to prevent concentration. The authors note that the formulation can be restructured to maximise total benefits under a fixed cost cap, aligning with audited annual spending envelopes, and approximate ecological dynamics iteratively by updating benefits between runs to reflect diminishing returns. The project extends this approach through a five-objective lexicographic framework and an explicit spatial dispersion penalty, ensuring allocation is not only ecologically efficient but also geographically balanced across Pará's municipalities.

In summary, the literature explored different areas of conservation planning, such as site selection, biodiversity prioritisation, and enforcement targeting in the Amazon. These models either treat funding as a binary decision or rely on ratio-based objectives, making it hard to audit in a public policy context. This project builds upon the literature by proposing a MILP framework. It develops upon simple binary selection to include a continuous decision variable for restoration scale, exploring the key trade-offs within the problem to optimise the allocation of scarce financial capital to target key municipalities requiring reforestation efforts.

3 Problem Description

3.1 The Problem

The decision-making problem addressed involves optimising the allocation of an annual budget of USD \$489,850,000 [Section 5.1.1], consisting of both internal and external funding, for reforestation across the 144 municipalities in Pará. The model developed in this report recognises five main ecological objectives (derived in Section 5.1.2), that determine how to optimally allocate funding between the municipalities:

- (i) **Biodiversity richness** is an important factor in discerning between potential reforestation sites, as areas with greater species richness are generally more resilient and ecologically productive.
- (ii) To combat species population declines in Pará, threatened species lists are used as key policy measures to prioritise conservation actions and guide resource allocation. Hence, this report defines a **biodiversity threat** metric based on the concentration of species classified under the IUCN Red List [27].
- (iii) **Carbon sequestration** serves as an indicator for identifying optimal reforestation sites, as areas with tree species that are more efficient at sequestering carbon contribute to greater overall carbon absorption.
- (iv) SEMAS also aims to prioritise municipalities in Pará that have been subject to intense levels of deforestation in recent times; hence **deforestation urgency** will be considered when distinguishing between funding allocation. SEMAS defines urgency by analysing net deforestation levels faced by municipalities in the last 5 years [28], representing current pressure while smoothing volatility.
- (v) To capture the risk that restoration efforts may not be sustained, this report constructs a **reversal risk** index that reflects persistent deforestation pressure within each municipality
- (vi) Finally, the model incorporates a **spatial objective**, ensuring that funding is spread throughout Pará. This is done to ensure fairness in the optimal solution and ensure funding reaches diverse biomes and species instead of being overly concentrated in one area.

This report does not attempt to identify specific land parcels within each municipality; instead, it determines the recommended municipality-level scale of restoration for SEMAS to support, leaving micro-siting to subsequent technical planning by implementers. The model is intended as an annual prioritisation tool recommending, for each municipality, whether it should be funded and at what scale in the current planning cycle. The model is designed to be re-run annually using updated indicators and data, allowing SEMAS to adopt a dynamic and adaptive reallocation approach while maintaining a static optimisation structure in each run.

Given limited resources and the immense scope of deforested land in Pará, it is not feasible to restore all eligible degraded land within the programme timeframe of 1-year, leading to SEMAS facing recurring trade-offs. The objectives set out by SEMAS are not perfectly aligned and often conflict with one another. For example, municipalities with high carbon sequestration potential may not coincide with those with the

greatest biodiversity richness or the highest concentration of threatened species. Similarly, areas facing urgent deforestation pressures may also exhibit high reversal risk, meaning that investing in such locations could yield lower long-term ecological benefits. The decision-making problem, therefore, consists of determining how to balance these competing objectives under a fixed budget and time constraint. As such, this report seeks to provide SEMAS, the decision maker, with a clear outline of what municipalities to prioritise, and to what extent, using a **Multiple-Objective Linear Programming** approach.

3.2 Dealing with Uncertainty

The deterministic problem described previously [Section 3.1] assumes a fully known SEMAS annual budget at the time of decision-making. In practice, SEMAS's budget of USD \$489,850,000 aggregates from two structurally distinct sources that become available at fundamentally different points in the planning cycle [Section 5.1.1].

First, state-level funds consisting of PMV (USD \$2.17M), PSA (USD \$9.53M), and FEMA (USD \$27.84M) are confirmed at the start of the planning cycle, creating a certain fixed budget of USD \$39.54 million. In contrast, the remaining sources, consisting of international and private climate organisations, are subject to disbursements, delays, and fluctuations – and will normally become confirmed later in the year. **The difference in timing structures provides the natural foundation for a two-stage stochastic programme with simple recourse.** Before the uncertain budget is revealed, SEMAS must undertake a set of costly and irreversible preparation actions occurring when a municipality is chosen for restoration using the confirmed funds. These first-stage set-up activities consist of **ecological surveys, administrative and bureaucratic work, and establishing contractor work and other formal labour** [29]. **These activities take time, consume resources, and must happen before the final international budget disbursements are confirmed. Once the international budget is confirmed, SEMAS makes second-stage decisions on which set-up municipalities actually receive restoration funding, at what scale, and whether surplus budget permits emergency activation of additional municipalities.**

This two-stage structure is non-trivial for several reasons. **Set-up costs** are sunk as they are paid in full, from the initial fixed budget, regardless of external funding, meaning if a municipality is chosen and set up, but the stage 2 restoration budget is insufficient to fund the minimum viable scale, the set-up cost is entirely wasted. **This penalty is what creates the incentive for conservative stage 1 portfolio sizing without requiring a separately calibrated penalty parameter.** Furthermore, a chosen municipality cannot be deselected in stage 2. Finally, surplus budget in stage 2 can only be deployed to municipalities that were either planned in stage 1 or emergency-activated at a higher cost, creating an asymmetry between under-committing and over-committing in stage 1.

3.3 Limitations and Assumptions

This study makes explicit modelling assumptions to align with the data and SEMAS's decision authority. Firstly, the model is annual and static within-year; it does not simulate ecological transitions or long-run habitat recovery dynamics. It is instead designed for annual re-optimisation using updated inputs. Moreover, the framework assumes that higher deforestation, biodiversity, or urgency scores translate proportionally into greater ecological returns from investment.

A key limitation of the model is that it operates at the municipal level rather than at the parcel or landscape scale. While this aligns with Pará’s governance structure, it introduces aggregation bias, as ecological conditions vary significantly within municipalities. High-priority restoration areas may be averaged out, and funded municipalities may still contain land unsuitable for restoration. Because micro-siting is left to later implementation stages, ecological effectiveness ultimately depends on local planning capacity, reducing the spatial precision of the optimisation.

Furthermore, a limitation of the biodiversity density parameter is potential underreporting of species occurrences, particularly in remote or less accessible municipalities. Since observation data is usually collected through field surveys, municipalities with limited monitoring infrastructure may exhibit lower recorded biodiversity levels [30] and be underrepresented in the mathematical model. Finally, the report assumes that travel time to cities fully captures the variation in restoration delivery costs, potentially overlooking other factors such as land conditions, terrain type, and labour availability. Additionally, applying a baseline accessibility cost may overlook within-municipality variations.

Despite these limitations, the model is designed to distribute the available budget fairly while balancing the objectives stated above, thereby providing strategic insight into how SEMAS can optimally allocate funding to combat deforestation. Future research could address these limitations through parcel-level optimisation, probabilistic species density modelling, and a more granular cost structure.

4 Mathematical Formulation

4.1 Deterministic Model

The deterministic model is formulated as an MILP solved using a lexicographic multi-objective method, solving objectives sequentially in a priority order, according to the policy set out by SEMAS. A sensitivity analysis was implemented to determine proportional tolerances to be implemented for each objective function using the estimated Utopian and Nadir points [Appendix A.1.2]:

4.1.1 Sets

- $i \in I$: Set of candidate municipalities in Pará, $|I| = 144$
- $j \in J$: Duplicate set of I $|J| = 144$ (used to define distances between two municipalities: $d_{i,j}$)

4.1.2 Parameters

- Bio_i : Normalised biodiversity richness score for municipality i , reliability-weighted $\in [0, 1]$.
- $\tilde{E}_i$: Normalised biodiversity threat score for municipality i , weighted by IUCN conservation status $\in [0, 1]$.
- $\tilde{C}ar_i$: Normalised carbon sequestration score for municipality i , weighted by wood density $\in [0, 1]$.
- $\tilde{U}_i$: Normalised urgency index for municipality i , mean annual deforestation 2020–2024 $\in [0, 1]$.
- $\tilde{R}ev_i$: Normalised reversal risk index for municipality i , share of years 2015–2024 exceeding 4,000 ha $\in [0, 1]$.

- A_i : Maximum eligible hectares available for restoration in municipality i (ha).
- D_i : Total cumulative deforested area in municipality i , 1986-2024 (ha).
- B : Annual budget envelope available to SEMAS (\$USD), estimated at \$489.85 million [Section 5.1.1].
- c_i : Per-hectare restoration cost in municipality i , accessibility-adjusted (\$USD/ha).
- $d_{i,j}$: Pairwise distance between municipalities i and j .
- m : Minimum viable project scale: minimum hectares funded if municipality i is selected, set at $m = 200$ ha (defined by Restaura Amazônia programme [31])
- K : Annual implementation capacity: maximum total hectares fundable across all municipalities, set at $K = 565,000$ ha (defined by the PEAA and PRVN-PA)
- ε_M : Minimum number of municipalities required to receive funding per planning cycle, set at $\varepsilon_M = 25$ as baseline (derived from Amazon Fund's Green Municipalities Programme)
- α_B, β_B : Weights on biodiversity richness and threat in F_1 , with $\alpha + \beta = 1$.
- θ : maximum share of the annual budget that any single municipality may receive per planning cycle, set at $\theta = 0.20$ (based on previous BNDES [32] and Amazon Fund allocation [33])
- The construction of all normalised indices and the derivation of c_i, A_i , and D_i from raw data sources are detailed in Section 5.1. The scalar parameters B, K , and m are discussed below.

4.1.3 Decision Variables

- $x_i \geq 0$: Hectares funded for reforestation in municipality i
- $y_i \in \{0, 1\}$: Binary decision variable to 1 if municipality i receives funding, 0 otherwise
- ϕ : Maximum interaction-based impact
- $z_{i,j} \in \{0, 1\}$: Variable for linearisation: equals $y_i y_j$

4.1.4 Objective Functions

Mathematically, the model can be expressed by the following objective functions:

$$F_1 = \max \sum_{i=1}^{|I|} (\alpha_B B i o_i + \beta_B \tilde{E}_i) \cdot x_i \quad (1)$$

$$F_2 = \max \sum_{i \in I} \tilde{C} a r_i \cdot x_i \quad (2)$$

$$F_3 = \max \sum_{i \in I} \tilde{U}_i \cdot x_i \quad (3)$$

$$F_4 = \min \sum_{i \in I} \tilde{Rev}_i \cdot x_i \quad (4)$$

$$F_5 = \min \phi \quad (5)$$

Subject to:

$$\phi \geq \sum_{i=1}^{|I|-1} \sum_{j=i+1}^{|I|} z_{i,j} \varsigma_{i,j}, \text{ where } \varsigma_{i,j} = \begin{cases} \frac{1}{d_{i,j}}, & \text{if } \varsigma_{i,j} \leq 250 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$$\phi \geq 0 \quad (7)$$

$$z_{i,j} \geq y_i + y_j - 1 \quad \forall i < j \quad (8)$$

$$z_{i,j} \leq y_i, \quad z_{i,j} \leq y_j \quad \forall i < j \quad (9)$$

$$\sum_{i \in I} c_i \cdot x_i \leq B \quad (10)$$

$$x_i \leq A_i \cdot y_i \quad \forall i \in I \quad (11)$$

$$x_i \geq m \cdot y_i \quad \forall i \in I \quad (12)$$

$$\sum_{i \in I} x_i \leq K \quad (13)$$

$$\sum_{i \in I} y_i \geq \varepsilon_M \quad (14)$$

$$\sum_{i \in I_{arc}} c_i \cdot x_i \geq \$14,820,000 \quad (15)$$

$$c_i \cdot x_i \leq \theta \cdot B \quad \forall i \in I \quad (16)$$

$$F_j \geq F_j^* \quad j = 1, 2, 3 \quad (\text{applied sequentially when solving } F_{j+1}) \quad F_4 \leq F_4^* \quad (\text{applied when solving } F_5) \quad (17)$$

$$x_i \geq 0, \quad y_i \in \{0, 1\} \quad z_{i,j} \in \{0, 1\} \quad \forall i \in I \quad \forall j \in J \quad (18)$$

Objective Function Construction For the full breakdown and detail on the construction of the objective functions refer to the Report Specific Process outline [Appendix A.1.2].

4.1.5 Constraint Explanations

Constraint (6) - Spatial Penalty: Requires the maximum-interaction-based impact decision variable to be at least equal to the sum of penalties between all pairs of selected municipalities in one iteration. The decision variable $z_{i,j}$ ensures that only penalties between two selected municipalities are counted. Distances greater than 250 km are assumed to have a negligible impact and are therefore assigned zero penalty.

Constraints (8-9) - Linearization constraints Ensure that the auxiliary binary variable $z_{i,j}$ correctly represents the logical AND of y_i and y_j , i.e., $z_{i,j} = y_i \wedge y_j$.

Constraint (15) - Arc of Reforestation Minimum: From the budget construction, all money provided from co-financing from the CIF, World Bank and BNDES is specifically for the area of the Arc of Reforestation located in the Tocantins–Araguaia river basin [Appendix C.1]. This constraint ensures that this budget has to be spread across municipalities that contain areas of the Tocantins–Araguaia basin.

Constraint (17) - Lexicographic preservation: Encodes the lexicographic ordering. When solving stage $k + 1$, the optimal values $F_1^*, \dots, F_k^*$ obtained from prior stages are appended as lower bounds, preserving the gains of higher-priority objectives throughout the sequential solution procedure.

4.2 Stochastic Model

The stochastic model retains the constraint structure of the deterministic formulation but simplifies the objective to F_1 alone. Three components are omitted: first, the full lexicographic hierarchy, as the deterministic optimal bounds on $F_2 - F_5$ derived at budget B become infeasible under adverse realisations $B^\omega < B$, and relaxing them to scenario-dependent bounds adds complexity disproportionate to the model's scope. Second, the geographic dispersion objective, to replicate the $\binom{144}{2} = 10,296$ pairwise binary variables across N scenarios, renders the SAA unmanageable. Lastly, the arc of reforestation with minimum expenditure, since the \$14.82M allocation falls within the uncertainty component of B^ω , making the floor infeasible. The model instead focuses on which municipalities to commit using F_1 as the sole ecological criterion. The formulation is as follows:

4.2.1 Additional Sets

- $\omega \in \Omega$: set of budget scenarios, each representing a possible realisation of the uncertain international budget.

4.2.2 Additional Parameters

- B^{fixed} : confirmed annual budget obtained from the state-level sources, known with certainty at the start of the planning cycle $B^{fixed} = \$39.54M$.

- B^ω : second stage budget under scenario ω , revealed after stage 1 decisions are made; $\mathbb{E}[B^\omega] = \$450.31\text{M}$.
- p^ω : probability of scenario ω , with $\sum_{\omega \in \Omega} p^\omega = 1$.
- f_i : fixed setup cost for municipality i .
- γ : late premium multiplicative cost for both the **set-up cost** and the **per-hectare restoration cost** in stage 2 using surplus budget, demonstrating higher mobilisation and contracting costs of unplanned/short-noticed restoration. Since no published figures for delayed environmental set-up procedures are available, $\gamma = 1.25$ is used as a conservative baseline assumption.

Fixed setup cost f_i : USD\$85,805 per municipality obtained from the project value of Amazon Fund's Green Municipalities Programme in Pará [29]; R\$45,591,647.24 divided by the 100 municipalities that directly benefited and converted at an exchange rate of R\$5.31 per USD. This cost incorporates activities and procedures needed before restoration funding can begin; it is treated as a sunk expenditure once the municipality is selected.

4.2.3 Additional Decision Variables

- $y_i \in \{0, 1\}$: **first-stage** binary decision variable equal to 1 if municipality is chosen and paid to be set up in stage 1, 0 otherwise. This decision is made before B^ω is revealed and is binding and irreversible.
- $v_i^\omega \in \{0, 1\}$: **second-stage** binary variable equal to 1 if municipality i is activated late using surplus budget in stage 2 under scenario ω , where $y_i = 0$. This variable allows surplus budget to be deployed productively rather than left unspent.
- $x_i^\omega \geq 0$: **second-stage** continuous variable representing the real scale of hectares of restoration funded in municipality i under scenario ω .

4.2.4 Auxiliary Variables

- $s_i^\omega \geq 0$: auxiliary variable tracking hectares funded in municipality i in scenario ω only when $v_i^\omega = 1$; 0 otherwise. Introduced to linearise the bilinear product $x_i^\omega \cdot v_i^\omega$ in the budget constraint.
- $\delta_i^\omega \in \{0, 1\}$: **second-stage** binary variable equal to 1 if municipality i received any funding in scenario ω , 0 otherwise. The purpose is to enforce the minimum viable scale for all funded municipalities in stage 2, no matter when they were set up.

4.2.5 Full Two-Stage Stochastic Formulation

Stage 1: SEMAS chooses which municipalities to set up for restoration before the uncertain international budget is revealed. The first-stage problem maximises expected net ecological benefit across both stages:

$$\max_y \quad [-\sum_{i \in I} f_i \cdot y_i + \sum_{\omega \in \Omega} p^\omega \cdot Q(y, \omega)] \quad (19)$$

The stage 1 set-up costs $\sum_{i \in I} f_i \cdot y_i$, which are paid for with certainty from the known B^{fixed} , are subtracted from the second-stage benefit $Q(y, \omega)$. The expected stage 2 benefit is a probability-weighted average of stage 2 outcomes across all possible scenarios.

Subject to:

$$\sum_{i \in I} f_i \cdot y_i \leq B^{fixed} \quad (20)$$

Set-up budget constraint: The total spending on municipality set-up may not exceed the known budget of $B^{fixed} = \$39.54M$ as these funds are directly available at the start of the planning cycle.

$$\sum_{i \in I} y_i \geq \varepsilon_M \quad (21)$$

Geographic equity constraint: At least $\varepsilon_M = 25$ municipalities must be set up in stage 1, identical to the deterministic model.

$$y_i \in \{0, 1\} \quad \forall i \in I \quad (22)$$

Binary domain: A municipality set-up decision is a binary choice, either it is chosen to be set up or not.

Uncertainty Revealed: After the first stage, set-up decisions are made, and costs $\sum_{i \in I} f_i \cdot y_i$ are paid to chosen municipalities, the real international budget B^ω is confirmed and revealed. The budget available for actual restoration in stage 2 is then:

$$B_{restoration}^\omega = B^\omega - \sum_{i \in I} f_i \cdot y_i \quad (23)$$

The total restoration budget available in stage 2 is the international budget, less the total spent on setup paid in stage 1. The setup costs create a financial consequence of stage-1 decisions that are permanently sunk.

Stage 2: SEMAS maximises the ecological benefit while being aware of the set-up municipalities $\{i : y_i = 1\}$ and the final restoration budget $B_{restoration}^\omega$. The second stage value function can be defined as:

$$Q(y, \omega) = \max_{x^\omega, v^\omega, \delta_i^\omega} \sum_{i \in I} (\alpha_B Bio_i + \beta_B \tilde{E}_i) \cdot x_i^\omega \quad (24)$$

This equation is the total ecological benefit which will be achieved under scenario ω .

Subject to:

$$\sum_{i \in I} c_i \cdot x_i^\omega + (\gamma - 1) \sum_{i \in I} c_i \cdot s_i^\omega + \gamma \sum_{i \in I} f_i \cdot v_i^\omega \leq B_{restoration}^\omega \quad (25)$$

Restoration budget constraint: Term one $\sum_{i \in I} c_i \cdot x_i^\omega$ is the standard restoration cost, term two $(\gamma - 1) \sum_{i \in I} c_i \cdot s_i^\omega$ charges the per-hectare premium on delayed municipalities, where $s_i^\omega = x_i^\omega \cdot v_i^\omega$, raising

their effective cost to $\gamma \cdot c_i$ per hectare. Term three $\gamma \sum_{i \in I} f_i \cdot v_i^\omega$ applies the premium to delayed set-up costs. Since $\gamma > 1$, the planned stage 1 setup is always cheaper, preserving the incentive to prepare.

$$x_i^\omega \leq A_i \cdot (y_i + v_i^\omega) \quad \forall i \in I \quad (26)$$

Eligible area and linking constraint: The funded area for restoration in municipality i may not exceed the eligible degraded area A_i . Only municipalities that have been chosen and set up, either in stage 1 or 2, can receive funding for restoration

$$y_i + v_i^\omega \leq 1 \quad \forall i \in I \quad (27)$$

Mutual exclusivity: A municipality cannot be set up in both stage 1 and stage 2.

$$x_i^\omega \geq m \cdot \delta_i^\omega \quad \forall i \in I \quad (28)$$

$$x_i^\omega \leq A_i \cdot \delta_i^\omega \quad \forall i \in I \quad (29)$$

$$\delta_i^\omega \leq y_i + v_i^\omega \quad \forall i \in I \quad (30)$$

Minimum scale: δ_i^ω indicates if municipality i receives funding for restoration under scenario ω . The minimum scale for hectares restored, at least $m = 200$ ha [31], is enforced, applying to both stage 1 and stage 2 set-up municipalities. $x_i^\omega = 0$ is forced whenever a municipality does not receive funding; restorative funding only occurs if a municipality is set up in stage 1 ($y_i = 1$) or set up via delayed activation in stage 2 ($v_i^\omega = 1$). For municipalities set up in stage 1 where $\delta_i^\omega = 0$, the funded area is zero, and the setup cost is therefore wasted.

$$c_i \cdot x_i^\omega \leq \theta \cdot B_{restoration}^\omega \quad \forall i \in I \quad (31)$$

Equity constraint: The maximum allowable funding to any single municipality cannot exceed $\theta = 0.20$, which is 20% of the total budget available for restoration under scenario ω . This cap applies to $B_{restoration}^\omega$ rather than B^ω , because the set-up costs have been committed and paid to the selected municipalities; therefore, the cap applies only to the funds remaining after set-up costs have been paid.

$$\sum_{i \in I} x_i^\omega \leq K \quad (32)$$

Implementation capacity constraint: Total funded hectares across all municipalities in stage 2 cannot exceed $K = 565,000$ ha.

$$s_i^\omega \leq x_i^\omega \quad \forall i \in I \quad (33)$$

$$s_i^\omega \leq A_i \cdot v_i^\omega \quad \forall i \in I \quad (34)$$

$$s_i^\omega \geq x_i^\omega - A_i(1 - v_i^\omega) \quad \forall i \in I \quad (35)$$

$$s_i^\omega \geq 0 \quad \forall i \in I \quad (36)$$

Linearisation: Multiplying a binary variable v_i^ω by a continuous variable x_i^ω causes nonlinearity; s_i^ω must be introduced to allow the use of logic that otherwise would not be possible, and it keeps the model linear. The four constraints together enforce $s_i^\omega = x_i^\omega \cdot v_i^\omega$ at any solution. This ensures that premiums are charged correctly and appropriately.

$$x_i^\omega \geq 0, \quad v_i^\omega \in \{0, 1\}, \quad \delta_i^\omega \in \{0, 1\}, \quad s_i^\omega \geq 0 \quad \forall i \in I \quad (37)$$

Domain constraints

4.2.6 Scenario Generation

The uncertain portion of the budget has a nominal value of \$450.31M, a stratified approach is used on 16 annual disbursement data points sourced from the Amazon Fund Portfolio Report [34] covering 2010 to 2025. The report infers the distribution structure and probability weights from these records; however, it must be noted that this is an assumption the report makes that the distribution will follow that of the source of disbursement.

Scaling: To ensure $\mathbb{E}[B^\omega] = B_{nominal} = \$450.31M$, the 16 observable disbursements are scaled:

$$\kappa = B_{nominal} / \bar{d} = 450.31 / 36.625 = 12.30 \quad (38)$$

where $\bar{d} = 36.625M$ is the mean disbursement by Amazon Fund.

Stratum definition: The 16 scaled scenarios are sorted by size and split into three strata:

- **Stratum 1 - Low** [USD\$73.8M, USD\$332.0M] Probability weight $p_1 = 6/16 = 0.375$.
- **Stratum 2 - Medium** [USD\$418.0M, USD\$491.8M] Probability weight $p_2 = 6/16 = 0.375$.
- **Stratum 3 - High** [USD\$627.1M, USD\$885.3M] Probability weight $p_3 = 4/16 = 0.250$.

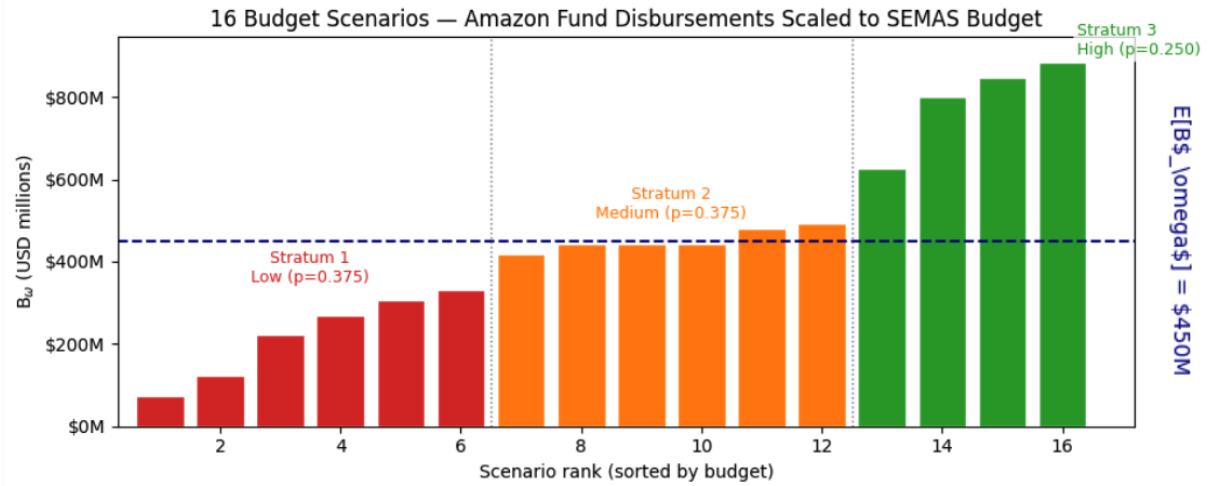


Figure 1: Scaled Budget Stratum Distributions [34]

Stratum allocation: The 6 / 6 / 4 divide, seen in Figure 1 is due to the natural distribution of the data being right-skewed. Furthermore, the gaps between the strata are due to the absence of historical observations between them; therefore, they are a feature of the empirical data used. Six observations fall in the lower range; six observations cluster near the expected budget; and four observations reach the high range of peak budget over-disbursements.

Within-stratum distribution: Within each stratum, B^ω is assumed to have a uniform distribution over the stratum interval $[a_k, b_k]$; this has been assumed due to the absence of historical resolution.

4.2.7 Stratified Sample Average Approximation and Convergence

Sample allocation: With total sample size N and weights $(p_1, p_2, p_3) = (0.375, 0.375, 0.250)$:

$$N_k = \lfloor p_k \cdot N \rfloor \quad k = 1, 2, 3 \quad (39)$$

Within stratum k , N_k independent draws from uniform $[a_k, b_k]$. The SAA objective:

$$\max_y \left[-\sum_{i \in I} f_i \cdot y_i + \sum_{k=1}^3 p_k \cdot \frac{1}{N_k} \sum_{\omega \in S_k} Q(y, \omega) \right] \quad (40)$$

Convergence analysis: To find the smallest sample size that produces a stable solution, the model is run at increasing values $N \in \{3, 10, 25, 50, 100, 200\}$. At each N , the model is solved $M = 20$ times using different random draws. The sample mean $\bar{z}_N$ and standard deviation $\hat{\sigma}_N$ for each N are recorded for each replication. Following Kleywegt et al. [35], convergence is assessed by comparing the upper bound $\bar{z}_N^{UB}$, the mean objective, against the lower bound $\bar{z}_N^{LB}$, obtained by fixing the best first stage solution $\bar{y}$ and comparing with an independent sample of $N' = 1000$ scenarios. Convergence is deemed achieved when the gap is negligible relative to the objective value, and the same N size is used for all reported results. The recourse problem value, RP, is reported as the mean objective $\bar{z}_N^{UB}$ across all $M = 20$ replications at

N^* , rather than a single seed solve. This design choice eliminates dependence on arbitrary seed choices and produces a statistically representative estimate consistent with Kleywegt [35].

4.2.8 Performance Metrics: EVPI and VSS

Two standard metrics will be used to quantify the usefulness and value of the stochastic formulation.

Expected Value of Perfect Information (EVPI):

$$EVPI = \mathbb{E}_\omega[WS(\omega)] - RP \quad (41)$$

EVPI measures how much SEMAS would and should be willing to pay to know the final budget B^ω before making any decisions. $\mathbb{E}_\omega[WS(\omega)]$ is the wait-and-see value - the expected objective under this perfect forecast. This optimally solves the model, as it knows the exact outcome under the real budget for each scenario in isolation. RP is the recourse problem value, meaning the objective achieved under uncertainty using the two-stage stochastic model.

Value of Stochastic Solution (VSS):

$$VSS = RP - EEV \quad (42)$$

VSS shows the benefit of using the stochastic model compared to the initial deterministic option regarding the budget. Expected value replaces B^ω with $\mathbb{E}[B^\omega] = \$450.31\text{M}$, solving the model as if it were certain. This makes the total available budget of $B^{fixed} + \mathbb{E}[B^\omega] = \489.85M , the same as solving the deterministic model at full expected budget. Stage 1 solution $\bar{y}$ is fixed and evaluated across all N SAA scenarios to compute EEV. VSS is therefore the ecological benefit of accounting for budget uncertainty over naively planning with the expected budget. By construction, $EEV \leq RP \leq \mathbb{E}_\omega[WS(\omega)]$, confirming $VSS \geq 0$ and $EVPI \geq 0$.

5 Computational Results

5.1 Data Description and Analysis

5.1.1 SEMAS's Budget Determination

SEMAS's budget (B) is derived from multiple sources, including state and programme funds, concessional climate finance from multilateral institutions, and private and blended restoration funds. External climate finance is assumed to be disbursed uniformly over five years (t), reflecting typical large-scale finance schedules. This report identifies 6 distinct capital sources:

$$B = \underbrace{B_{\text{PMV}}}_{\$2.17\text{M}} + \underbrace{B_{\text{PSA}}}_{\$9.53\text{M}} + \underbrace{(B_{\text{FEMA}} \times \phi_{\text{env}} \times \rho_{\text{FEMA}})}_{\$27.84\text{M}} + \underbrace{\frac{B_{\text{NPC}} \times \alpha_{\text{NPC}}}{t}}_{\$14.82\text{M}} + \underbrace{\frac{B_{\text{BNDES}} \times \alpha_{\text{PEAA}}}{t}}_{\$85.49\text{M}} + \underbrace{(B_{\text{TFFF}} \times h_P)}_{\$350.00\text{M}} = \$489.85 \text{ Million} \quad (43)$$

- **Programa Municípios Verdes (PMV):** A state programme strengthening municipal environmental management, historically allocating USD \$2.17 million (B_{PMV}) annually [29], which this report assumes remains unchanged in 2026.
- **Valoriza Territórios Sustentáveis (PSA):** A state fund that pays farmers for ecological services, including reforestation [36]. Funded and replenished annually to USD \$9.53 million (B_{PSA}) through Banpará bank's dividends. [37].
- **Fundo Estadual de Meio Ambiente (FEMA):** Pará's state environmental fund, replenished annually by state mining/water taxes. In 2026, FEMA's fund value is expected to be R\$1 billion (B_{FEMA}) [38]. In 2024, only 20% (ϕ_{env}) of FEMA spending went to environmental projects, with only 73% (ρ_{FEMA}) of those funds deployed [39]. Assuming the same execution rate and allocation proportion in 2026, FEMA contributes USD \$27.84 million to SEMAS's budget.
- **Nature, People and Climate (NPC) for Brazil:** The Climate Investment Funds (CIF) approved a USD \$47 million first-loss loan for Brazil's Arc of Restoration initiative in the Tocantins–Araguaia basin [Appendix C.1] [40]. This concessional capital has mobilised approximately USD \$200 million in co-financing from the World Bank and the Brazilian Development Bank (BNDES), creating a blended finance pool of USD \$247 million (B_{NPC}) [41]. Pará's allocation is assumed proportional to its basin coverage; with approximately 30% of the Tocantins–Araguaia basin located in the state [42, 43], it receives 30% of the funds (α_{NPC}), equivalent to USD \$74.10 million. Annualised, this results in an NPC contribution of USD \$14.82 million.
- **Public Call for Climate Mitigation Initiative:** Through its investment arm BNDESPAR, BNDES has committed USD \$95.33 million to two nature-based solutions (NBS) investment vehicles [32]: the Pátria LatAm Reforest Fund I [44] and Mombak's Amazon Reforestation Fund II [45]. These funds prioritise native forest regeneration within the Amazon Restoration Arc, including areas in Pará. This capital is expected to mobilise an additional USD \$718.78 million in private market co-investments, resulting in a total investment pool of approximately USD \$909.44 million (B_{CIF})

[32]. With the PEAA target of 47% (α_{PEAA}) of Amazon reforestation, SEMAS is assumed to be proportionally allocated USD \$427.44 million (B_{BNDES}), contributing approximately USD \$85.49 million annually.

- **Tropical Forests Forever Facility (TFFF):** Brazil introduced the TFFF ahead of COP30, establishing a USD \$125 billion endowment for annual payments for preserved forest areas [46]. At the proposed rate based on the proposed payment structure of USD \$4 per hectare (B_{TFFF}), Pará's 87.5 million hectares (h_P) of forest [47] could generate USD \$350 million in annual dividends for SEMAS to allocate. This estimate assumes the endowment is fully fund However, this figure is contingent on the USD \$125 billion endowment being fulfilled, and payments are being fully deployable.

5.1.2 Mathematical Derivation

Biodiversity Richness Index Data was sourced from SiBBr [48], containing georeferenced species occurrence observations. Records were filtered for valid geographic coordinates, location, country, and state. Then each record was spatially joined to the 144 municipality polygons. Let S be the set of species prevalent in Pará, $s \in \{1, \dots, |S|\}$, and $O_{i,s}$ be the set of observed species in each municipality i , $o_{i,s} \in \{1, \dots, |O_{i,s}|\}$. Then the *species richness* (r_i) of municipality i can be defined as the number of unique species s observed in municipality i : $R_i = |\{s \in S : s \text{ observed in municipality } i\}|$. To obtain an accurate biodiversity richness index, species richness was first normalised ($\hat{R}_i$):

$$R_{min} = \min_{j \in I} R_j \text{ and } R_{max} = \max_{j \in I} R_j \quad (44)$$

$$\rightarrow \tilde{R}_i = \frac{R_i - R_{min}}{R_{max} - R_{min}} \quad (45)$$

To account for fairness in the biodiversity richness index, a weight w_i was calculated. The exponential saturation function was deemed an appropriate weighting method, as it penalises municipalities with sparse sampling. A scaling parameter τ , and the cardinality of observed species in each municipality ($n_i = |o_i|$) was used to derive the weight:

$$w_i = 1 - \exp\left(\frac{-n_i}{\tau}\right) \quad (46)$$

Where, τ is calibrated so that the municipality at the median record count receives $w_i = 0.5$:

$$\tau = \frac{n_{\text{median}}}{\ln(2)} \quad (47)$$

Resulting in:

$$Bio_i = w_i \times R_i, \quad \forall i \in I \quad (48)$$

Biodiversity Threat Index To create a biodiversity threat index, species observation data from SiBBr and conservation status data from the IUCN Red List were used [49]. Let T_l be the set of threatened species in

Pará, $t_l \in \{1, \dots, |T_l|\}$, $T \subseteq S$, where l is the animal's corresponding conservation status, $l \in \{VU, EN, CR\}$ - vulnerable, endangered and critically endangered respectively. A weight w_l is defined for each conservation status according to the percentage decline of the species in the last 10 years. For example, critically endangered species have seen a population decline of 80% in the last 10 years:

$$w_l = \begin{cases} 0.3, & l = VU \\ 0.5, & l = EN \\ 0.8, & l = CR \end{cases} \quad (49)$$

A binary variable $p_{t,i}$ indicating whether species t is present in municipality i is used to derive a biodiversity extinction risk for each municipality i (E_i):

$$E_i = \sum_{t=t_l}^{|T_l|} w_l \times p_{t,i} \quad (50)$$

This value was normalised similarly to Equation 45.

Carbon Sequestration Index Both data from the Global Wood Density Database v.2 [50] and species observation from SiBBr were used to derive a carbon sequestration index. Since the mass of carbon in trees is directly correlated to the biomass of their wood [51], wood density values for species present in Pará were taken to calculate carbon sequestration metrics for each municipality. Let C be the set of species capable of performing carbon sequestration - species of class *Magnoliopsida*, *Liliopsida*, *Pinolnopsida*, $c \in \{1, \dots, |C|\}$, $C \subseteq S$. A weight w_c is defined for each genus of species c , according to its wood density, which is used as a proxy for carbon sequestration. Species observation data from SiBBr were used to calculate the proportion of different tree genera present in municipalities. Given the set $O_{i,s}$, an accurate carbon sequestration value can hence be computed as follows:

$$Car_i = \sum_{c \in C_i} \frac{o_{i,s}}{\sum_{j \in C_i} o_{i,j}} \times w_c \quad (51)$$

This value was normalised similarly to Equation 45.

Urgency Index As described in Section 3.1, the urgency index measures recent deforestation pressure by capturing which municipalities are experiencing the most contemporary active forest loss. For each municipality i , the raw urgency is the 5-year mean of annual deforestation D_i over 2020–2024:

$$U_i = \frac{1}{5} \sum_{t=2020}^{2024} D_{i,t}$$

This value was normalised similarly to Equation 45.

Risk of Reversal Index As stated in Section 3.1, reversal risk measures the persistence of high deforestation activity in each municipality over the past decade, serving as a proxy for the likelihood that newly restored forest would be cleared again. A binary variable $b_{i,t}$ is used to indicate whether a municipality has exceeded the benchmark of 4,000 ha set by the PMV [29]. Furthermore, the binary

variable $b_{i,t}$ prevents penalising municipalities that are actively improving, by examining the yearly deforestation levels relative to the previous year:

$$\Delta D_{i,t} = D_{i,t} - D_{i,t-1} \quad (52)$$

$$b_{i,t} = \begin{cases} 1, & \text{if } D_{i,t} \geq 4000 \text{ and } \Delta D_{i,t} \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (53)$$

Leading to:

$$Rev_i = \frac{1}{10} \sum_{t=2015}^{2024} b_{i,t} \quad (54)$$

This value was normalised similarly to Equation 45.

Travel Time and Restoration Cost Parameter Calculation Reforestation costs vary across municipalities due to differences in accessibility, which affect the transport of materials, labour power, and logistical overhead. This is captured using the Malaria Atlas Project (MAP) “Travel time to cities” raster [52]. This source provides the estimated travel time, in minutes, from each pixel to the nearest urban centre with population $\geq 50,000$ [53].

Zonal statistics were computed in QGIS, where for each municipality polygon, the median pixel value of the MAP raster was extracted from IBGE [54], related to all municipalities in Para, producing TT_i in minutes. The median was chosen over the mean to reduce the influence of outlier pixels in large, sparsely populated municipalities. A remoteness index was then computed using min-max normalisation:

$$remote_i = \frac{TT_i - \min_j TT_j}{\max_j TT_j - \min_j TT_j}$$

The per-hectare restoration cost is computed as:

$$c_i = 2041.27 + 728 \times remote_i \quad (\text{USD} / \text{ha})$$

Where the cost of \$2041.27 & \$728 is grounded in empirical Brazil restoration literature [16].

Eligible Restoration Area The eligible restoration area is the total land in each municipality that has experienced deforestation between 1986 and 2024, making it suitable for reforestation efforts. Data was sourced from the MapBiomas Collection 10 Deforestation & Secondary Vegetation asset [55], accessed via Google Earth Engine (GEE):

$$A_i = \sum_{t=1986}^{2024} D_{i,t}$$

Spatial Data To create a spatial parameter, the distances between two municipalities $d_{i,j}$ s.t. $i \neq j$ were computed. Centroids for each municipality were calculated based on border data sourced from IBGE.

5.1.3 Index Evaluation

This report runs correlation tests across all parameters to ensure accuracy and avoid the fallacy of double-counting and using redundant data:

Biodiversity Metrics The overarching objective of maximising biodiversity consists of a species richness index (Bio_i) and a species threat index ($\tilde{E}_i$). Since both are based on animal observation data from the SiBBr database, and may have underlying patterns that could lead to double counting, a Pearson's correlation test was run to determine how best to incorporate both parameters. The results yielded an r value of $r = 0.82$, indicating a significant correlation between both parameters, yet not a direct one-to-one. As a result, both indices were combined into a singular objective function containing a composite index to avoid redundancy while preserving both underlying objectives. Hence:

$$\max \sum_{i=1}^{|I|} (\alpha_B Bio_i + \beta_B \tilde{E}_i) x_i \quad (55)$$

Furthermore, since the carbon sequestration index ($\tilde{Car}_i$) uses tree species as a proxy, and $T \subseteq S$, multicollinearity was also tested between Bio_i and $\tilde{Car}_i$. Here, the Pearson correlation test yielded $r = 0.02$, indicating no correlation. Hence, both indices can safely be used as separate objective functions without leading to double counting.

5.2 Policy Payoffs

Table 2 presents the multi-objective payoff table, constructed from five single-objective optimisation runs, which records all objective values for each solution. The diagonal entries represent the utopia values [Appendix A.1]. The off-diagonal entries reveal the trade-offs among the remaining objectives when each is pursued in isolation.

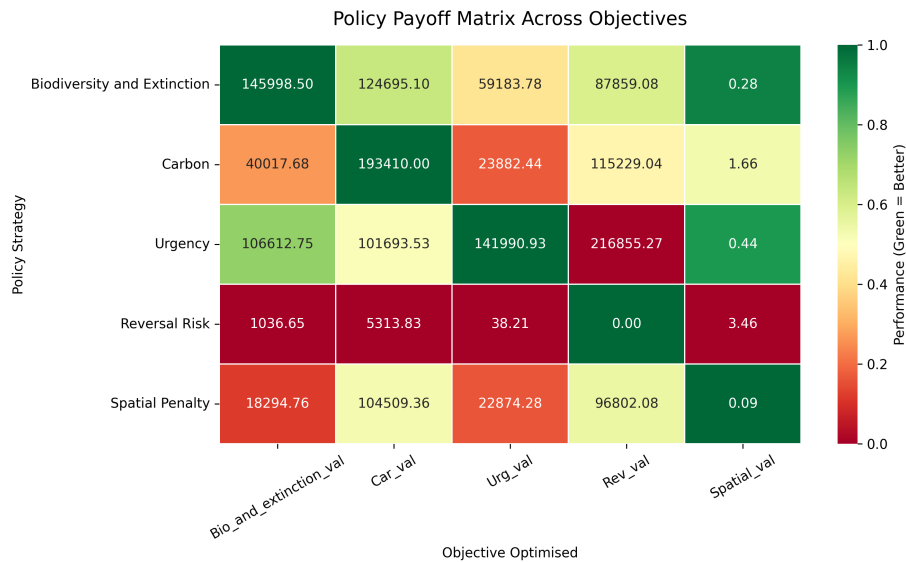


Figure 2: Policy Payoff Matrix

The results indicate that no single-objective policy yields satisfactory outcomes across all dimensions. Reversal risk presents the most striking finding: when it is minimised in isolation, every ecological objective collapses by orders of magnitude. Biodiversity falls from its utopia of 145,998.50 to just 1,036.65; urgency drops to 38.21, and carbon sequestration to 5,313. In practical terms, municipalities that are safest in terms of investment risk offer almost no restorative value - a policy focused solely on securing returns would fund restoration in entirely the wrong places.

An equally important conflict arises in the Urgency row. Maximising urgency yields the worst reversal risk score among all policies (216,855), exposing a fundamental trade-off between acting where reforestation is most needed and where it is most likely to endure. This finding directly motivates placing reversal risk as the fourth priority in the lexicographic sequence [Appendix A.1.2], ensuring urgency gains are preserved before the riskiest allocations are reconsidered.

Biodiversity and carbon sequestration are naturally aligned. When biodiversity alone is optimised, carbon sequestration still achieves 124,695.10, 64.47% of its individual best, without being directly targeted. This suggests that high-biodiversity municipalities also tend to possess meaningful carbon sequestration potential, partially justifying their placement at first and second in the objective hierarchy.

Payoff Radar Charts: Individual Optimisation



Figure 3: Payoff Radar Charts

The radar charts in Figure 3 visually support these observations, with each polygon's shape revealing the trade-offs incurred when a single objective is pursued in isolation. The reversal risk plot is the most extreme case, collapsing to a single line as all other objectives fall to their worst recorded values. The urgency chart performs strongly across biodiversity, carbon and spatial penalty but collapses on reversal risk, reflecting the trade-off already established. Carbon and spatial penalty charts show similarly asymmetric polygons. The biodiversity chart is the exception, producing the most uniform profile and maintaining reasonable performance across all objectives without directly targeting them.

Collectively, the asymmetric polygons across the five charts confirm that no single objective policy can adequately address the full range of priorities considered in this report, providing strong motivation for the relaxed lexicographical approach adopted.

5.3 Deterministic Results

The deterministic model aims to determine the optimal financial allocation of the total budget of USD \$489,850,000 available to SEMAS. This report utilised PuLP, implementing the Gurobi optimisation solver.

5.3.1 Evaluating the Optimisation Framework

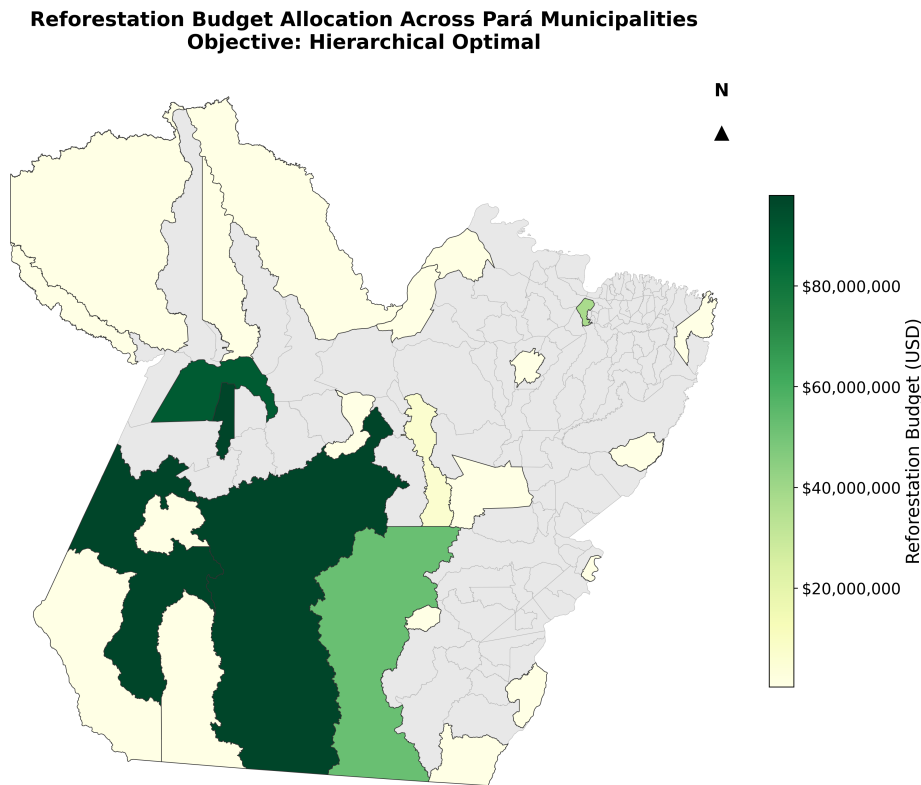


Figure 4: Lexicographic Allocation Map Plot

As seen in Figure 4, the hierarchical MILP heavily prioritises **Belterra**, **Santarém**, and **Altamira**, allocating the maximum allowable funds to each (Refer to Table 15 in Appendix B). Specifically, the equity constraint is strictly binding for these three municipalities, pushing them to the $\theta = 0.20$ budget cap limit. Maximising these variables to their absolute upper bounds mathematically indicates that these municipalities offer the highest composite ecological yield under the lexicographic hierarchy. To satisfy minimum intervention and spatial equity constraints (Equation 12), the model allocates a long tail of exact 200-hectare minimums to regions like **Viseu**, **Itupiranga**, and **Ulianópolis**. This establishes a foundational conservation presence across Pará without draining capital from higher-yield priority zones.

5.3.2 Heuristic Comparison

To quantify the added benefits from the mathematical model, it was benchmarked against a realistic heuristic allocation. Emulating historical Brazilian Government environmental policies for resource allocation, being historically driven by simple criteria rankings [25], where municipalities included in the

"priority list" were selected based on a retrospective analysis and specific threshold criteria, including total deforestation area and increase in deforestation area. This heuristic ranks municipalities by recent deforestation levels (urgency U_i) and allocates top-down funding until all resources are exhausted, while satisfying the political global constraints to ensure a realistic policy decision (Refer Appendix A.3 for algorithm and B 13 for allocations).

Table 1: Comparison of Total Objective Function Utilities: Proposed MILP Policy vs. Heuristic Selection

Objective	Description	MILP Optimal Value	Heuristic Value	Relative Gap (%)
F_1 (Max)	Biodiversity & Extinction	140,747.23	106,680.14	24.20%
F_2 (Max)	Carbon Sequestration	127,860.74	101,686.33	20.47%
F_3 (Max)	Intervention Urgency	90,556.79	141,988.58	-56.80%*
F_4 (Min)	Reversal Risk	127,082.92	216,865.49	70.65%
F_5 (Min)	Spatial Penalty (ϕ)	0.1083	0.4257	293.07%

*A negative gap indicates the heuristic achieved a strictly higher value for this specific objective.

Evaluating this heuristic through the formal objective functions exposes its systemic risk. While the heuristic successfully captures 56.8% more intervention urgency (F_3) by acting greedily, it incurs unacceptable optimality gaps across all other metrics. Most notably, the heuristic leaves 24.20% of the optimal biodiversity utility completely un-captured (F_1), while simultaneously inflating the Reversal Risk Exposure (F_4) by 75.83% compared to the risk-hedged MILP. Furthermore, without a spatial constraint, the heuristic allocates investments randomly, resulting in a Spatial Dispersion Penalty (F_5) that is 293.07% worse than the MILP's cohesive ecological corridors. The comparative radar plots (5) geometrically reinforce the danger of a one-dimensional, urgency-driven policy; the heuristic's narrow, unbalanced footprint formally confirms that it is highly Pareto-inefficient, lacking the multi-objective resilience required by SEMAS. Figure 6 shows the heuristic's budget allocation within Pará.

**MILP vs. Heuristic: Normalized Objective Performance
(Common Payoff Table Extremes)**

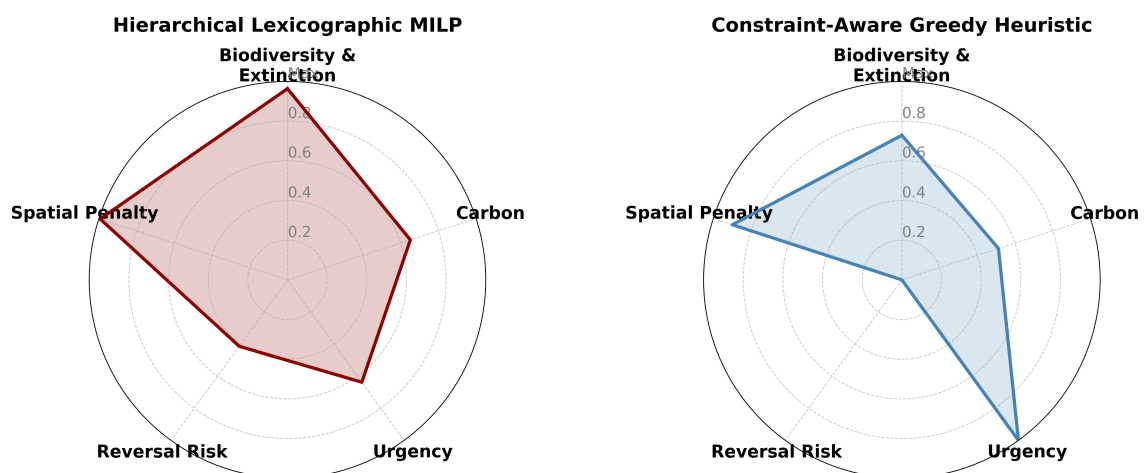


Figure 5: MILP (left) vs Heuristic Radar Plot (right)

Reforestation Budget Allocation Across Pará Municipalities
Objective: Heuristic

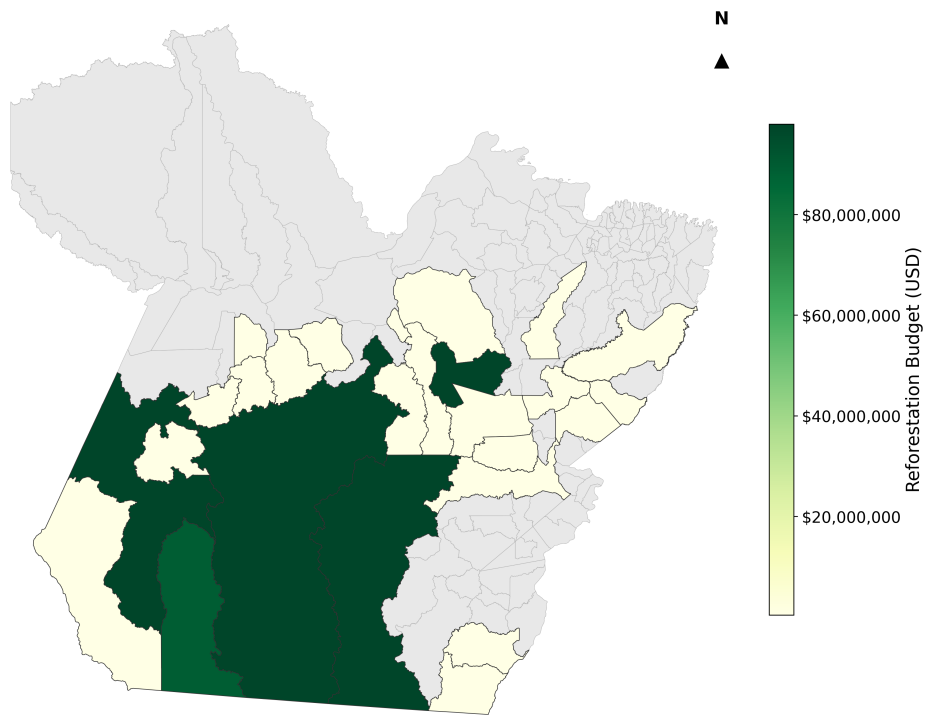


Figure 6: Heuristic Allocation Map Plot

5.4 Deterministic Sensitivity Analysis

5.4.1 Sensitivity Analysis: Fiscal Fluctuations

To conduct sensitivity analysis and test the robustness of the baseline optimisation framework, the deterministic model was re-solved under three empirically rounded budget scenarios: a low budget of US\$221.37 million, the baseline medium budget of US\$489.85 million, and a high budget of US\$790 million [Appendix A.2]. The results are summarised in Table 2 and illustrated in Figure 7 (refer B 14 15 16 for allocation tables).

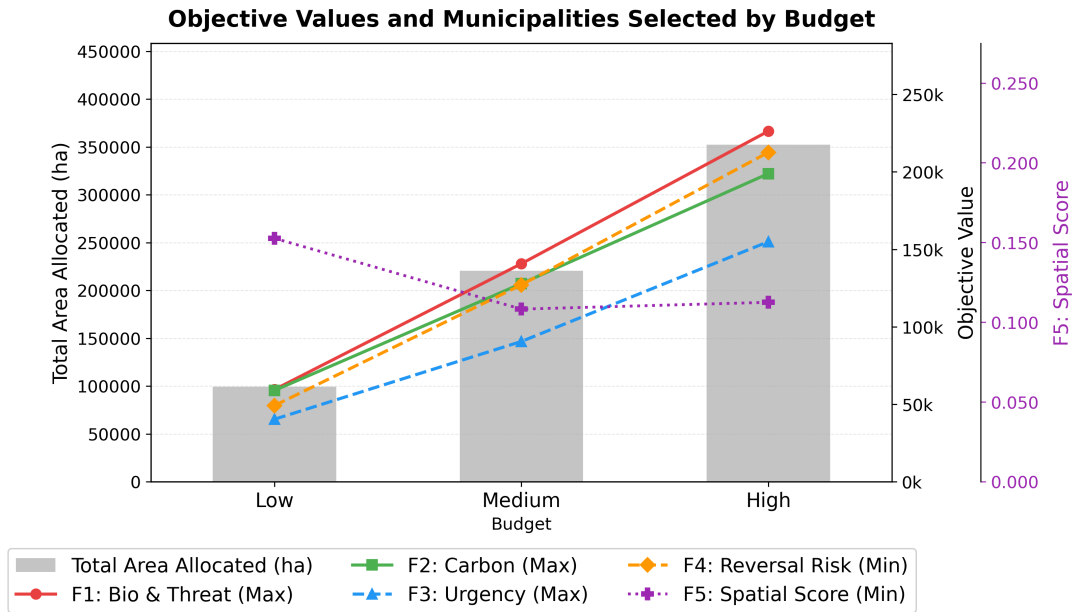


Figure 7: Objective Values under Budget Sensitivity Analysis

Analysis indicates that total restoration area scales linearly with the budget (up to 3.57× for a 3.55× funding increase), expanding from 99,347 ha (Low) to 352,473 ha (High), with the average per-hectare cost remaining stable at approximately US\$2,230/ha (fluctuating less than 1%). This demonstrates that the model can absorb additional financial resources without incurring cost inflation or diminishing returns. Furthermore, the model consistently selects exactly 25 municipalities across all scenarios, prioritising deeper investment into already-selected priority municipalities with high ecological benefits, rather than expanding the portfolio of funded municipalities.

Objectives F_1, F_2, F_3 showcase roughly linear improvements aligned with the 3.57x budget increase from low to high budget scenarios. Spatial dispersion penalty F_5 substantially improves from low to medium budget scenarios (Low=0.153, medium=0.106), reflecting significantly better geographical coverage. The reversal risk (F_4) increases disproportionately by 432%, outpacing both the budget increase of 357%. This is primarily driven by the single-municipality budget cap of $\theta = 0.20$. At low budget levels, this cap restricts allocations mainly in low-risk, high-yield municipalities. But as the budget increases, those safe municipalities hit their 20% funding cap. The solver is mathematically forced to push the excess capital into riskier, volatile municipalities (like **Santarém** expanding from 0 to 73,232 ha) to keep spending. Figure 8 compares the budget allocations between the low and high budget scenarios.

Table 2: Sensitivity Analysis of Budget Allocation Scenarios

Scenario	Total Area Allocated (ha)	Municipalities Selected	Avg. Cost per Hectare (US\$)
Low	99,347	25	2,228.26
Medium	220,673	25	2,219.80
High	352,473	25	2,241.31

Comparison of Reforestation Budget Allocations Across Pará Municipalities

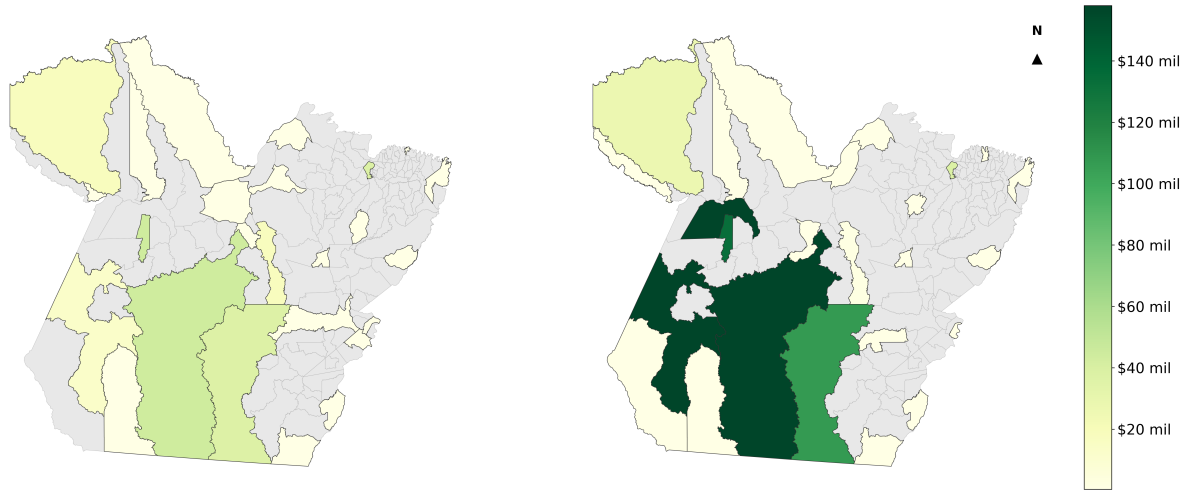


Figure 8: Low budget Scenario (left) vs High Budget Scenario (right)

5.4.2 Sensitivity Analysis: Policy Changes

To validate the robustness of the proposed baseline policy, a "Risk-Averse Investor" structural sensitivity analysis is conducted by re-ordering the lexicographic hierarchy. Reversal Risk (F_1), Biodiversity (F_2), Urgency (F_3), and Carbon (F_4), maintaining the 5% tolerance ($\alpha = 0.05$). This simulates operating under extreme public scrutiny, where preventing the failure of funded projects is the priority objective.

Evaluating reversal risk as the primary objective successfully reduces portfolio risk exposure by 92.78% [Table 3]. However, this incurs a severe opportunity cost: urgency degrades by 92.95% and biodiversity by 24.00%, visually confirmed from the radar plot comparison in Figure 9. This exposes a fundamental spatial conflict in Pará—active deforestation frontiers inherently carry high reversal risks. Forced to build a "safe" portfolio, the solver abandons volatile zones, stripping **Altamira** and **Itaituba** of primary funding, visually verified from the comparison of Figure 10 and Figure 4. By avoiding the active frontier, the model marginally improves cost-efficiency, expanding the total restored area by 2.35% (to 225,858.8 ha).

Ultimately, this demonstrates that strict risk aversion fundamentally alters ecological outcomes. To restore ecologically rich, active frontiers, SEMAS must be willing to accept a higher risk of project reversal.

Table 3: Objective Value Comparison: MILP (Lexicographic) vs. Risk-Adverse

Metric	Objective	Original Optimal Policy	Risk-Adverse Policy	Relative Change (%)
Reversal Risk	Minimise	127,082.92	9,171.38	-92.78%
Biodiversity	Maximise	140,747.23	106,963.20	-24.00%
Urgency	Maximise	90,556.79	6,383.82	-92.95%
Carbon	Maximise	127,860.74	121,030.88	-5.34%
Spatial Score	Minimise	0.1083	0.1500	+38.50%

**MILP vs. Risk-Adverse Investor: Normalised Objective Performance
(Common Payoff Table Extremes)**

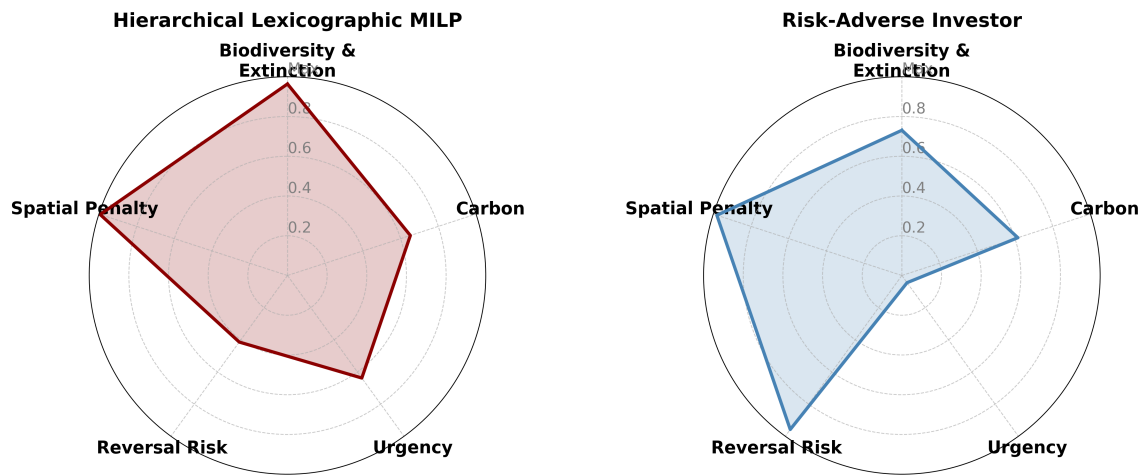


Figure 9: Original Optimisation (left) vs Risk-Adverse Optimal Allocation (right)

Reforestation Budget Allocation Across Pará Municipalities
Objective: Risk-Averse Investor

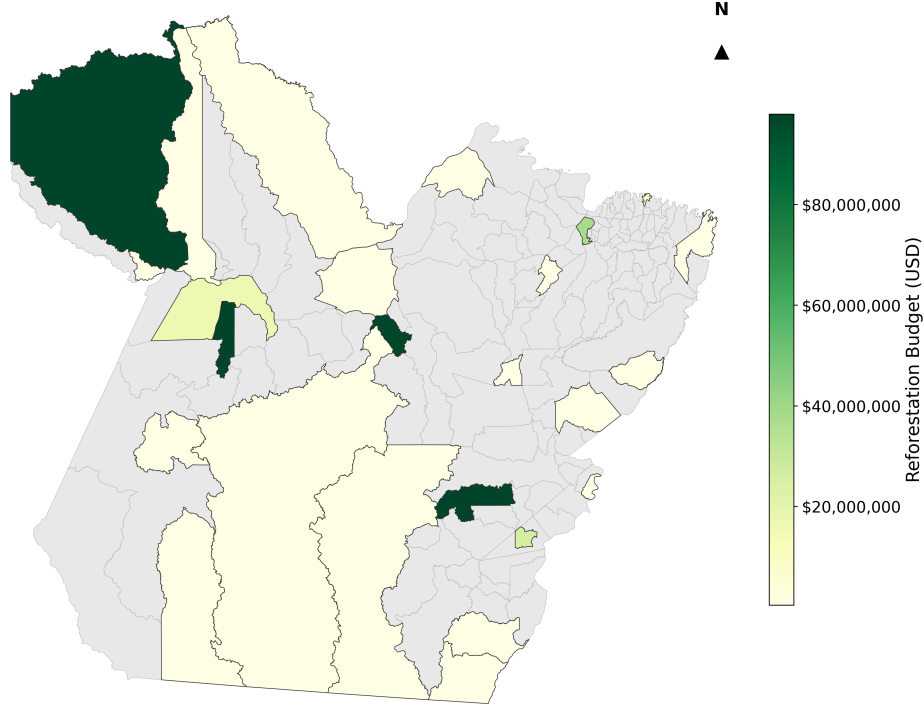


Figure 10: Risk-Adverse Policy Allocation Map Plot

5.5 Stochastic Results

Table 4: Results for each N tested in Stochastic Model

N	UB	LB	Gap	Gap%
3	127,978	130,732	-2,754	-2.15%
10	130,286	131,005	-719	-0.55%
25	128,904	130,606	-1,701	-1.32%
50	129,652	130,672	-1,020	-0.79%
100	130,606	130,078	+529	+0.41%

Convergence: Table 4 and Figure 11 shows the stratified SAA model was run across $N \in \{3, 10, 25, 50, 100, 200\}$ different scenarios and replicated 20 times each time. Notice that at smaller sample sizes, the gap is negative; this is due to the SAA upper bound underestimating the true optimal value. This occurs due to too few scenarios per replication, causing the mean objective across the 20 replications to be a poor estimate of the true RP due to high variance across replications. The lower bound captures the single best solution found and evaluates it against $N' = 1000$ representative scenarios, which produces a more accurate estimate. This negative gap is a recognised feature of smaller sample SAA [35]. However, at $N = 100$, the gap turns positive at 529 biodiversity-weighted hectares, representing 0.405% of the upper bound objective, with the coefficient of variation stabilising at 0.0200. This is consistent with the

convergence criterion described in Section 4.2.7, and therefore $N^* = 100$ is used for all reported results and $N = 200$ was not required, given convergence was already achieved.

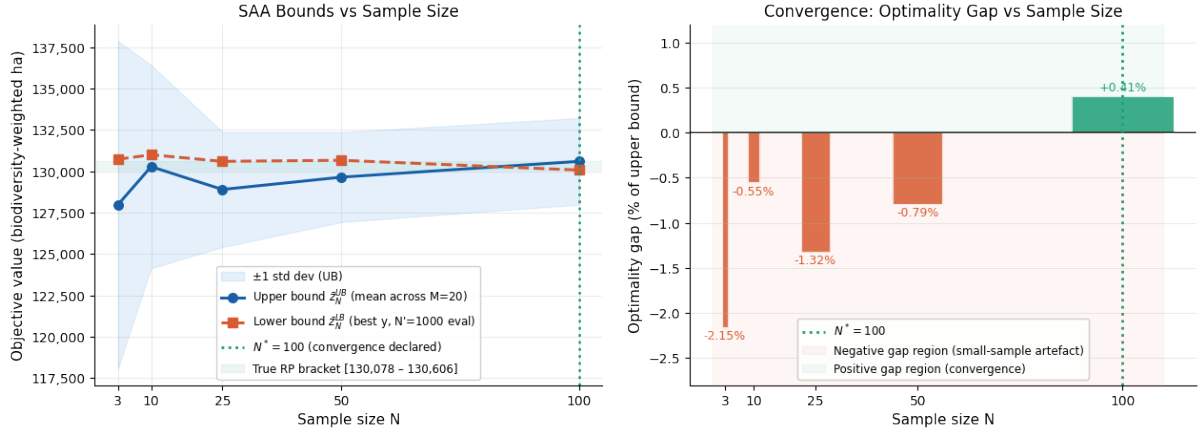


Figure 11: Stochastic Convergence Plot Analysis

Table 5: Results at $N^* = 100$

Metric	Value	% of RP
EEV	130,471.39 bio-ha	99.897%
RP	130,606.39 bio-ha	100%
$\mathbb{E}[WS]$	130,608.65 bio-ha	100.002%
VSS	135.00 bio-ha	0.103%
EVPI	2.26 bio-ha	0.002%

Performance metrics: Table 5 show the stochastic model produces a result of $RP = 130,606.39$ biodiversity-weighted hectares reported as the mean objective across $M = 20$ replications. This shows that if SEMAS follows the stochastic model’s stage 1 recommendations, it can expect to deliver that on average across all possible budget outcomes. It is important to note that the stochastic model does not outperform the deterministic model in every scenario; however, it delivers greater expected ecological benefit across the full distribution of budget outcomes, with $VSS = 135.00$ biodiversity-weighted hectares, confirming the value of accounting for an uncertain budget in stage 1 planning. An EVPI of 2.26 bio-ha (0.002% of RP) indicates that perfect budgeting information adds minimal value, suggesting that SEMAS should focus on operational recourse capacity rather than spending money on forecasting potential budgets. The rule of $EEV \leq RP \leq \mathbb{E}[WS(\omega)]$ holds, confirming $VSS \geq 0$ and $EVPI \geq 0$.

Comparison: Tables 6 and 7 show the stochastic model chooses municipalities that are lower in cost per hectare (\$2,182 versus \$2,247) and average reversal risk (0.108 versus 0.420), meaning they remain fundable under low budget scenarios. Both models chose 25 municipalities and share 8 municipalities in common. The deterministic model incurs \$1,716,100 in setup costs at an expected budget of \$450.31M, because the restoration budget is insufficient to fund them at a minimum of 200 ha. The stochastic model

commits to cheaper municipalities at the same expected budget, with a mean waste of \$1,630,295, a reduction of \$85,805 (equal to one municipality's set-up cost). This illustrates that the stochastic model's advantage lies not in eliminating wasted setup costs but in selecting municipalities with lower per-hectare costs and lower reversal risk, ensuring that funded municipalities deliver reliable restoration.

Table 6: Deterministic Portfolio: 25 Selected Municipalities

Municipality	Bio	Cost/ha (\$)	Rev.	Ha Funded	Budget (\$)
Belterra	0.5901	2,085	0.0	46,989.6	97,970,000
Itaituba	0.5520	2,287	1.0	42,829.5	97,970,000
Santarém	0.6072	2,158	0.4	41,853.2	90,298,977
Altamira	1.0000	2,414	1.0	40,589.4	97,970,000
São Félix do Xingu	0.4670	2,280	1.0	22,968.9	52,358,803
Belém	0.6182	2,047	0.0	18,564.5	38,002,519
Anapu	0.1941	2,177	0.9	3,015.5	6,564,630
Afuá	0.0254	2,160	0.0	462.9	1,000,095
Trairão	0.1048	2,148	0.8	200.0	429,557
Conceição do Araguaia	0.1522	2,076	0.3	200.0	415,250
Ulianópolis	0.0133	2,077	0.4	200.0	415,355
Cametá	0.0473	2,097	0.0	200.0	419,405
Viseu	0.0742	2,098	0.3	200.0	419,511
Tucumã	0.0087	2,099	0.0	200.0	419,826
Santana do Araguaia	0.0768	2,102	0.5	200.0	420,405
Novo Repartimento	0.2102	2,107	1.0	200.0	421,457
Jacareacanga	0.4103	2,489	0.8	200.0	497,834
Brasil Novo	0.0570	2,153	0.2	200.0	430,557
Novo Progresso	0.4294	2,206	1.0	200.0	441,235
Gurupá	0.0619	2,295	0.0	200.0	458,909
Almeirim	0.5178	2,544	0.4	200.0	508,827
Oriximiná	0.5759	2,569	0.0	200.0	513,877
Alenquer	0.0907	2,672	0.5	200.0	534,339
Faro	0.0110	2,769	0.0	200.0	553,854
Palestina do Pará	0.0191	2,074	0.0	200.0	414,777
Total / Avg	0.277	2,247	0.420	220,673.5	489,849,999

Table 7: Stochastic Portfolio: 25 Selected Municipalities (Stage-2 Allocation at $\mathbb{E}[B^\omega] = \$450.31\text{M}$)

Municipality	Bio	Cost/ha (\$)	Rev.	Ha Funded	Budget (\$)
Parauapebas	0.5405	2,121	0.1	42,255.3	89,632,975
Santarém	0.6072	2,158	0.4	41,544.5	89,632,975
Belterra	0.5901	2,085	0.0	42,990.9	89,632,975
Altamira	1.0000	2,414	1.0	37,135.3	89,632,975
Itaituba	0.5520	2,287	1.0	22,571.2	51,630,456
Belém	0.6182	2,047	0.0	18,564.5	38,002,519
<i>19 municipalities set up, unfunded at $\mathbb{E}[B^\omega]$</i>				0.0	0
Total / Avg	0.186	2,182	0.108	205,061.7	448,164,875

Bio, Cost/ha, and Rev. averages in the total row are across all 25 selected municipalities.

The 19 unfunded municipalities were set up in stage 1 at \$85,805 each, serving as reserve capacity for low-budget scenarios.

Table 8: Portfolio Comparison Summary

Metric	Deterministic	Stochastic
Municipalities selected	25	25
Municipalities in common	8	
Total hectares funded	220,673.5	205,061.7
Total budget spent (\$)	489,849,999	448,164,875
Avg biodiversity composite	0.277	0.186
Avg reversal risk	0.420	0.108
Avg cost per hectare (\$)	2,247	2,182
Mean wasted setup cost (\$)	1,716,100	1,630,295



6 Conclusion

This study develops a multi-objective mixed integer linear programming framework to optimise the allocation of SEMAS's budget in order to fund reforestation efforts at the municipality level in Pará, Brazil. The framework incorporates the competing objectives of biodiversity, extinction risk, carbon sequestration, the urgency of deforestation, and spatial dispersion. These objectives are managed through relaxed lexicographical ordering, aligned with SEMAS's priorities under SDG 15.2.

The deterministic results confirm that the framework successfully directs capital to ecologically productive municipalities. It identified Belterra, Santarém, and Altamira as offering the highest return on investment, with each of these municipalities receiving maximum allowable funds. The deterministic frameworks' optimal portfolio achieved a mean biodiversity score of 0.33, against a global municipal average of 0.14, while targeting municipalities with urgency indices 161% above baseline. This shows the model can direct funds to critical zones with strong ecological value. When compared with a heuristic-driven model that mirrors Brazil's historical urgency-driven policy, this model captured 24.20% more biodiversity utility. It also reduced reversal risk exposure by 70.65%. Its only drawback was in targeting areas in urgent need of intervention. These results show that this report's deterministic framework is Pareto-superior to existing practices, highlighting the benefits of optimised fund allocation for reforestation in Pará.

The stochastic extension validates the robustness of the deterministic findings under budget uncertainty. An EVPI of 2.26 biodiversity-weighted hectares confirms that perfect budget forecasting adds negligible value. SEMAS should therefore prioritise operational recourse capacity rather than improved budget forecasting accuracy. Critically, the VSS of 135 biodiversity-weighted hectares (0.103% of the RP value) demonstrates that while the stochastic model is theoretically superior, the practical ecological gain over the deterministic approach is negligible. Given that the deterministic model is substantially less computationally demanding while accommodating the full lexicographic objective hierarchy and delivering near-identical ecological outcomes, SEMAS should adopt the deterministic framework as the standard for annual planning, reserving stochastic analysis for scenario stress-testing rather than operational decision-making.

This study contains several limitations that warrant acknowledgement. Firstly, under the medium budget scenario [Section 5.4.1] and estimated per-hectare restoration costs, the model only reforests 220,673 hectares of forest, rising to 385,850 hectares under the high budget scenario. At these reforestation rates, SEMAS would fail to meet the 5.65 million-hectare reforestation target in 5 years, instead requiring between 14.6 and 25.6 years. Therefore, while the model is effective, SEMAS will require substantially more funding to meet its 2030 targets. To meet the restoration target within the 5 years, the minimum budget required using this optimisation model was calculated at US\$ 11.67 billion [Appendix A.4], showcasing the scale of investment required when implementing the optimisation model. Regarding methodological limitations, the municipality-level granularity introduces aggregation bias, as ecological conditions vary substantially within municipalities. Additionally, the methods behind biodiversity reporting in remote areas may systematically disadvantage well-preserved but under-monitored regions. Furthermore, the

model's static, 1-year planning horizon does not capture long-run ecological dynamics or multi-year restoration trajectories. Further research could address these limitations through parcel-level optimisation, dynamic multi-period planning and by determining biodiversity through probabilistic species density modelling.

Nonetheless, this study provides SEMAS with a prioritisation tool that balances ecological impact and fiscal accountability, demonstrating how mathematical programming can make environmental policy actionable under SEMAS's PEAA commitment.

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Appendix

A Processes and Techniques

A.1 Lexicographic Optimisation Process

A.1.1 Overview

The model employs a relaxed lexicographical (Hierarchical) method where the competing objectives are optimised in the sequence of their relative priority. Once an optimal value is found for a higher-priority objective, it is converted into a preserving constraint with an associated tolerance factor to bound subsequent optimisations. These preserving constraints enforce the hierarchy between objectives by preventing the optimisation of lower-level objectives from interfering with higher-level ones. The tolerance factors are included in order to relax the preserving constraints, softening the strict priority structure between objectives by allowing controlled deviations from the optimal value of higher-priority objectives. This added flexibility enables lower-priority objectives to meaningfully influence the solution, resulting in outcomes that more holistically reflect the trade-offs inherent in the problem.

A.1.2 Report Specific Process

For this report, the tolerance factors (Δ_i) were determined as values proportional to the estimated Utopian and Nadir points:

- **Utopian Point:** A hypothetical best-case scenario where each objective achieves its individual optimum simultaneously. Formally, for each objective, F_i find its optimal value F_i^* , then the Utopian point is $(F_1^*, F_2^*, \dots, F_n^*)$. The Utopian is infeasible due to trade-offs among our objectives, which means that improving one objective degrades another.
- **Nadir Point:** The Nadir point represents the worst values of each objective among the Pareto-optimal solutions. Formally, examine all Pareto-efficient solutions, and for each objective F_i , take the worst value it attains on that front. As this is computationally intractable, this report estimates the Nadir points from the payoff table (i.e., the worst value each objective takes across the single-objective runs).

Together, the Utopian and Nadir points define a range of each objective:

$$\text{Range of } F_i = |F_i^{\text{Nadir}} - F_i^{\text{Utopian}}| \quad (56)$$

The Tolerances of each objective are then defined as

$$\Delta_i = \varpi \cdot (\text{Range of } F_i), \varpi \in [0, 1] \quad (57)$$

Where ϖ controls the degree of permitted degradation relative to each objective's spread, ensuring that the tolerances are consistent.

For this report, a sensitivity analysis was conducted across $\varpi \in \{0.05, 0.10, 0.15\}$, re-solving the full five-step lexicographic sequence under each setting. A uniform value of $\varpi = 0.05$ was selected, balancing solution flexibility with objective preservation. Table 9 details the tolerance values calculated for each objective.

Objective	Priority	Range of F_i	Δ_i
Biodiversity & Extinction Risk (F_1)	1st	144961.85	7248.09
Carbon Sequestration (F_2)	2nd	188096.18	9404.81
Urgency (F_3)	3rd	141952.72	7097.64
Reversal Risk (F_4)	4th	216855.27	10842.76

Table 9: Lexicographic tolerance ranges, and absolute tolerances for each objective.

(i) **Maximise Biodiversity (F_1):**

$$\max F_1 = \sum_{i=1}^{|I|} (\alpha_B Bio_i + \beta_B \tilde{E}_i) \cdot x_i \quad (58)$$

Let F_1^* be the optimal value. To allow for carbon sequestration potential, the preserving constraint is relaxed with Δ_1 :

$$\sum_{i=1}^{|I|} (\alpha_B Bio_i + \beta_B \tilde{E}_i) \cdot x_i \geq F_1^* - \Delta_1 \quad (59)$$

(ii) **Maximise Carbon Sequestration (F_2):**

$$\max F_2 = \sum_{i \in I} \tilde{C}ar_i \cdot x_i \quad (60)$$

Let F_2^* be the optimal value. To allow for the selection of municipalities in urgent need of reforestation, the preserving constraint is relaxed with Δ_2 :

$$\sum_{i \in I} \tilde{C}ar_i \cdot x_i \geq F_2^* - \Delta_2 \quad (61)$$

(iii) **Maximise Targeting Municipalities in Urgent need of reforestation(F_3):**

$$\max F_3 = \sum_{i \in I} \tilde{U}_i \cdot x_i \quad (62)$$

Let F_3^* be the optimal value. To allow for the least-risky selection of municipalities in terms of their risk of reforestation effort reversal, the preserving constraint is relaxed with Δ_3 :

$$\sum_{i \in I} \tilde{U}_i \cdot x_i \geq F_3^* - \Delta_3 \quad (63)$$

(iv) **Minimise Reversal Risk (F_4):**

$$\min F_4 = \sum_{i \in I} \tilde{R}ev_i \cdot x_i \quad (64)$$

Let F_4^* be the optimal value. To allow for the spatial distribution of municipalities to be considered, the preserving constraint is relaxed with Δ_4 . As this is a minimisation, the $F_4^* + \Delta_4$ acts as an upper bound.

$$\sum_{i \in I} \tilde{Rev}_i \cdot x_i \leq F_4^* + \Delta_4 \quad (65)$$

(v) **Minimise Spatial Connectivity (F_5):**

Finally, we minimise the variable ϕ to introduce fairness in the optimal solution by ensuring that funding reaches across different diverse biomes and species, instead of being overly concentrated in one area. As this is the last objective to be optimised, no preserving constraint or tolerance is needed

$$\min Z_5 = \phi \quad (66)$$

A.2 Budget Scenario Determination for Sensitivity Analysis of Deterministic Model

In order to determine the budget scenarios for the sensitivity analysis of the deterministic model, this report used the budget of USD \$489.85 million (Section 5.1.1) as the standard condition. For the low and high budget conditions, this report took the mean value of the data points within the intervals (Table 10, Table 11) for the low and high conditions detailed in Stratum 1 and 3 in Section 4.2.6. These low budget ($\hat{B}_{\text{low}}$) and high budget ($\hat{B}_{\text{high}}$) were determined as follows:

$$\hat{B}_{\text{low}} = \frac{73.8 + 332.0 + 307.4 + 270.5 + 221.5 + 123.0}{6} = \text{USD } \$221.37 \text{ Million} \quad (67)$$

$$\hat{B}_{\text{high}} = \frac{799.2 + 885.3 + 627.1 + 848.4}{4} = \text{USD } \$790 \text{ Million} \quad (68)$$

Table 10: Low Conditions ($k = 1$)

Year	B_t (USD M)
2010	73.8
2019	332.0
2020	307.4
2021	270.5
2022	221.3
2023	123.0

Table 11: High Conditions ($k = 3$)

Year	B_t (USD M)
2014	799.2
2017	885.3
2018	627.1
2025	848.4

A.3 Heuristic Baseline Policy Algorithm

The heuristic simulates a human planner who satisfies hard institutional constraints sequentially before greedily deploying the remaining budget. Municipalities are ranked in descending order of urgency score $\tilde{U}_i$ and proceed through three sequential phases.

Phase 1 — Arc of Deforestation Compliance The heuristic first directs spending to Arc municipalities until the \$12.35 million basin minimum is met, working down the urgency-ranked list and applying the same per-step limits as subsequent phases.

Phase 2 — Geographic Equity Compliance If fewer than $\varepsilon_M = 25$ municipalities have been funded, the heuristic iterates through remaining unfunded municipalities in urgency order, allocating exactly $m = 200$ ha to each until the equity floor is satisfied.

Phase 3 — Greedy Fill The residual budget is deployed greedily in urgency order, scaling each municipality to its effective area ceiling $A_i^{\text{cap}} = \min(A_i, \frac{\theta B}{c_i})$ until the global budget or implementation capacity is exhausted.

At every allocation step across all three phases, five limits are applied simultaneously:

$$x_i^{\text{add}} = \min \left(x_i^{\text{target}} - x_i, \frac{B_{\text{rem}}}{c_i}, \frac{\theta B - \text{Cost}_i}{c_i}, K_{\text{rem}}, A_i^{\text{cap}} - x_i \right) \quad (69)$$

At every allocation step, the hectares added to municipality i are bounded by five simultaneous limits:

1. The gap to the current phase target ($x_i^{\text{target}} - x_i$)
2. The globally remaining budget ($\frac{B_{\text{rem}}}{c_i}$)
3. The residual per-municipality budget cap θB ($\frac{\theta B - \text{Cost}_i}{c_i}$)
4. The remaining implementation capacity K (K_{rem})
5. The eligible land ceiling A_i^{cap} ($A_i^{\text{cap}} - x_i$)

A.4 Minimum Budget Required to Meet 5.65 Million Hectare Restoration Goal

To quantify the absolute minimum financial investment required to meet the 5 year policy target of restoring 5.65 million hectares, the original lexicographic MOLP model was reformulated into a single-objective cost-minimisation framework. Spatial connectivity constraints were relaxed to establish a pure theoretical financial baseline without geographic dispersion penalties, while still implementing physical bound constraints. The model implementation was formulated:

Objective Function

Minimise the total cost of reforestation across all selected municipalities:

$$\min Z = \sum_{i \in I} c_i \cdot x_i \quad (70)$$

Constraints

Target Area Constraint: Ensures the 5-year target of 5.65 million hectares is met or exceeded.

$$\sum_{i \in I} x_i \geq 5,650,000 \quad (71)$$

Geographic Equity: Ensures that at least the minimum required number of municipalities ($N_{\min} = 25$) receive funding.

$$\sum_{i \in I} y_i \geq 25 \quad (72)$$

Arc of Deforestation Minimum: Ensures that the 5-year scaled minimum investment is directed toward the highly vulnerable Arc of Deforestation municipalities.

$$\sum_{i \in I_{arc}} c_i \cdot x_i \geq 74,100,000 \quad (73)$$

Dynamic Budget Share Limit: Ensures no single municipality absorbs more than a set threshold ($\theta = 20\%$) of the total allocated budget, preventing monopolisation of funds.

$$c_i \cdot x_i \leq \theta \sum_{j \in I} c_j \cdot x_j \quad \forall i \in I \quad (74)$$

Eligible Land Limit: A logical bound ensuring the restored area does not exceed the physically available eligible land in any selected municipality.

$$x_i \leq A_i \cdot y_i \quad \forall i \in I \quad (75)$$

Minimum Viable Scale: A logical bound ensuring that if a municipality is selected for funding, it receives at least the minimum operational project scale ($m = 200$ ha).

$$x_i \geq m \cdot y_i \quad \forall i \in I \quad (76)$$

Domain Bounds: Defines the continuous nature of the area allocated and the binary nature of municipality selection.

$$x_i \geq 0, \quad y_i \in \{0, 1\} \quad \forall i \in I \quad (77)$$

B Allocation Tables

B.1 Hierarchical Allocation Table

Table 12: Allocation by Municipality: Hierarchical Optimal

Municipality	Area (ha)	Cost (USD \$)
Itaituba	42829.47	97970000.00
Altamira	40589.37	97970000.00
Belterra	46989.61	97970000.00
Santarém	41853.16	90298977.07
São Félix do Xingu	22968.89	52358803.35
Belém	18564.47	38002519.28
Anapu	3015.47	6564630.22
Afuá	462.92	1000094.85
Faro	200.00	553854.00
Alenquer	200.00	534338.97
Oriximiná	200.00	513877.12
Almeirim	200.00	508827.41
Jacareacanga	200.00	497833.77
Gurupá	200.00	458908.91
Novo Progresso	200.00	441234.92
Brasil Novo	200.00	430556.89
Trairão	200.00	429557.47
Novo Repartimento	200.00	421456.89
Santana do Araguaia	200.00	420404.87
Tucumã	200.00	419826.25
Viseu	200.00	419510.65
Cametá	200.00	419405.45
Ulianópolis	200.00	415355.16
Conceição do Araguaia	200.00	415249.95
Palestina do Pará	200.00	414776.54

Table 13: Allocation by Municipality: Heuristic Method

Municipality	Area (ha)	Cost (USD \$)
Pacajá	46671.69	97970000.00
São Félix do Xingu	42977.72	97970000.00
Itaituba	42829.47	97970000.00
Altamira	40589.37	97970000.00
Novo Progresso	40517.24	89388108.44
Novo Repartimento	200.00	421456.89
Portel	200.00	455069.03
Marabá	200.00	417143.60
Uruará	200.00	423929.14
Anapu	200.00	435396.20
Itupiranga	200.00	418563.83
Senador José Porfírio	200.00	450755.73
Paragominas	200.00	420825.68
Placas	200.00	424244.75
Moju	200.00	415723.36
Jacareacanga	200.00	497833.77
Rondon do Pará	200.00	418826.83
Rurópolis	200.00	424086.95
Trairão	200.00	429557.47
Medicilândia	200.00	438552.27
Santa Maria das Barreiras	200.00	418090.42
Mojuí dos Campos	200.00	417774.81
Dom Eliseu	200.00	416670.18
Santana do Araguaia	200.00	420404.87
Goianésia do Pará	200.00	416985.79

B.2 Heuristic Allocation Table

B.3 Budget Sensitivity Analysis Allocations

B.3.1 Low-Budget Scenario

Table 14: Low-Budget Allocation

Municipality	Area (ha)	Cost (USD \$)
Belterra	21235.26	44274000.00
Altamira	18342.90	44274000.00
Belém	18564.47	38002519.28
São Félix do Xingu	16632.93	37915647.16
Anapu	8252.97	17966558.99
Oriximiná	6970.77	17910602.63
Itaituba	5747.45	13146959.77
Alenquer	200.00	534338.97
Almeirim	200.00	508827.41
Porto de Moz	200.00	469113.54
Melgaço	200.00	467640.71
Terra Santa	200.00	455489.84
Novo Progresso	200.00	441234.92
Afuá	200.00	432082.32
Santana do Araguaia	200.00	420404.87
Viseu	200.00	419510.65
Tailândia	200.00	417432.90
Marabá	200.00	417143.60
Ulianópolis	200.00	415355.16
Conceição do Araguaia	200.00	415249.95
Tucuruí	200.00	414355.73
São Geraldo do Araguaia	200.00	414145.33
Salinópolis	200.00	412882.90
Brejo Grande do Araguaia	200.00	412672.50
Vitória do Xingu	200.00	411830.88

Table 15: Medium Budget Allocation

Municipality	Area (ha)	Cost (USD \$)
Itaituba	42829.47	97970000.00
Altamira	40589.37	97970000.00
Belterra	46989.61	97970000.00
Santarém	41853.16	90298977.07
São Félix do Xingu	22968.89	52358803.35
Belém	18564.47	38002519.28
Anapu	3015.47	6564630.22
Afuá	462.92	1000094.85
Faro	200.00	553854.00
Alenquer	200.00	534338.97
Oriximiná	200.00	513877.12
Almeirim	200.00	508827.41
Jacareacanga	200.00	497833.77
Gurupá	200.00	458908.91
Novo Progresso	200.00	441234.92
Brasil Novo	200.00	430556.89
Trairão	200.00	429557.47
Novo Repartimento	200.00	421456.89
Santana do Araguaia	200.00	420404.87
Tucumã	200.00	419826.25
Viseu	200.00	419510.65
Cametá	200.00	419405.45
Ulianópolis	200.00	415355.16
Conceição do Araguaia	200.00	415249.95
Palestina do Pará	200.00	414776.54

B.3.2 Medium-Budget Scenario

B.3.3 High-Budget Scenario

Table 16: High-Budget Allocation

Municipality	Area (ha)	Cost (USD \$)
Altamira	65460.04	158000000.00
Itaituba	69072.75	158000000.00
Santarém	73232.28	158000000.00
Belterra	64015.55	133467866.46
São Félix do Xingu	47064.54	107286130.13
Belém	18564.47	38002519.28
Oriximiná	10803.12	27757374.84
Anapu	860.35	1872969.91
Faro	200.00	553854.00
Alenquer	200.00	534338.97
Almeirim	200.00	508827.41
Jacareacanga	200.00	497833.77
Gurupá	200.00	458908.91
Novo Progresso	200.00	441234.92
Afuá	200.00	432082.32
Brasil Novo	200.00	430556.89
Parauapebas	200.00	424244.75
Santana do Araguaia	200.00	420404.87
Viseu	200.00	419510.65
Cametá	200.00	419405.45
Ulianópolis	200.00	415355.16
Conceição do Araguaia	200.00	415249.95
Palestina do Pará	200.00	414776.54
Tucuruí	200.00	414355.73
Maracanã	200.00	412199.09

B.4 Stochastic Results

B.5 Reversal Risk Allocation

Table 17: Alternative Allocation by Municipality: Prioritising Reversal Risk

Municipality	Area (ha)	Cost (USD \$)
Oriximiná	38129.74	97970000.00
Belterra	46989.61	97970000.00
Parauapebas	46185.60	97970000.00
Vitória do Xingu	47362.67	97527058.44
Belém	18564.47	38002519.28
Sapucaia	12535.98	25952001.38
Santarém	7632.06	16466303.03
Salinópolis	5058.68	10443204.48
Óbidos	200.00	537126.83
Almeirim	200.00	508827.41
Altamira	200.00	482737.24
Porto de Moz	200.00	469113.54
São Félix do Xingu	200.00	455910.65
Terra Santa	200.00	455489.84
Novo Progresso	200.00	441234.92
Afuá	200.00	432082.32
Trairão	200.00	429557.47
Santana do Araguaia	200.00	420404.87
Viseu	200.00	419510.65
Rondon do Pará	200.00	418826.83
Santa Maria das Barreiras	200.00	418090.42
Igarapé-Miri	200.00	415512.96
Ulianópolis	200.00	415355.16
Palestina do Pará	200.00	414776.54
Tucuruí	200.00	414355.73

C Auxiliary Maps and Figures

C.1 Tocantins–Araguaia Basin Municipalities

The Tocantins–Araguaia Basin is a major river system in central-northern Brazil that sits within the "Arc of Deforestation". The Arc of Deforestation is a frontier zone where forests are under immense deforestation pressure, largely due to agriculture and logging [56]. Figure 12 details the municipalities in Pará that either partially or fully lie in the Tocantins–Araguaia Basin, with the data sourced from [57].

Municipalities in the Tocantins–Araguaia Basin (Arc of Deforestation)

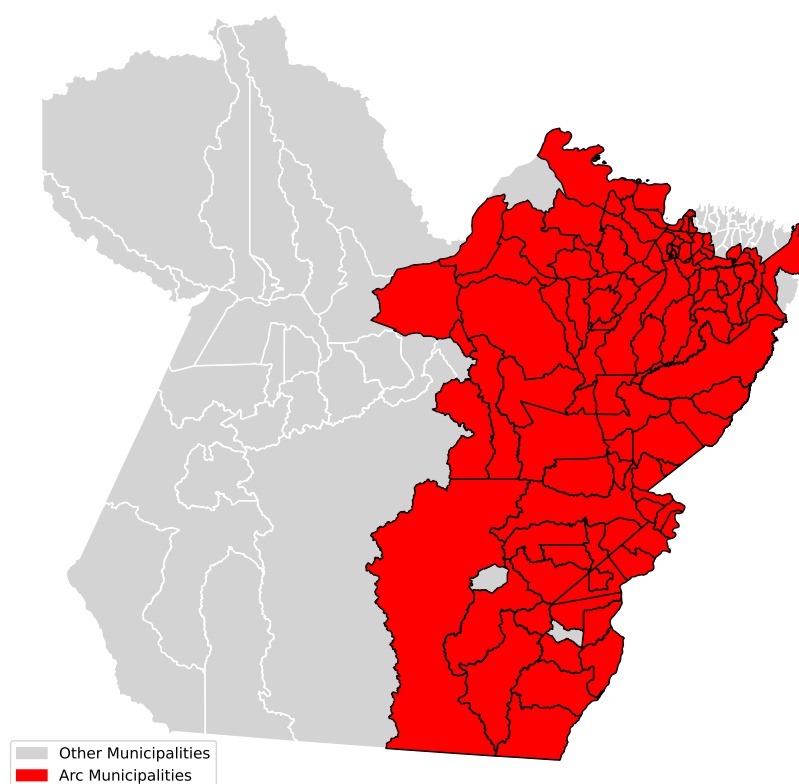


Figure 12: Municipalities in the Tocantins–Araguaia Basin (Arc of Deforestation), Pará State, Brazil. Red shading indicates municipalities falling within the basin, which represents the primary target area of the CIF Nature, People and Climate (NPC) Investment Plan. Grey shading indicates municipalities outside the basin.